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Aligning Large Language Models to the public: A background study and project proposal

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Abstract

Large Language Models (LLMs) are at the frontier of today's most capable AI systems. These models have shown remarkable capabilities to act for, and act with, humans in a variety of contexts. How these models behave is therefore crucially important, and they must act in alignment with how we would want them to behave. LLM alignment is a field that focuses on ensuring that these models' goals and behaviors are aligned to a target set of values and actions. Unfortunately, defining this target is non-trivial, and current methods often lack democratically grounded targets and robust evaluation methods. In this essay, I provide a background study of LLMs, alignment methods, AI safety, and human values. I then propose a project to research and implement a method to align LLMs to public values, along with strong evaluation methods to measure alignment success.

Contents

Chapter 1

Introduction

Globally, groups are building increasingly powerful AI systems. These systems are becoming embedded into many aspects of our lives and society and—notably—are operating with more autonomy than previous technologies. **Large Language Models** (**LLMs**) are a class of AI systems that have shown remarkable capabilities in understanding and generating natural language, audio and visual content. With these advanced capabilities, LLMs are being used to act both on our behalf and alongside us. The application of these systems motivates alignment between the systems and how we would want them to behave. Given the early stages of AI development, effective action and direction now could have outsized downstream effects.

In this report I focus on the **alignment** of LLMs; that is the process of taking a pre-trained LLM and making it behave more in a way that we want. I argue that current alignment methods lack democratically grounded processes and instead have perverse incentives that make alignment with public values/good unrealistic. I provide a project proposal to research and implement a method to align LLMs to public values and evaluate the success of this alignment in a way that is democratically grounded, transparent and fair.

1.1 The Alignment Process of LLMs

AI alignment is the field of study and practice that focuses on ensuring that AI systems' goals and behaviors are in line with what designers and end users intend. The field itself is relatively new, but draws on older ideas from control theory, ethics, policy, and governance.

LLMs are typically trained on vast amounts of text data using a **self-supervised learning** approach where they learn to predict the next word in a sentence. LLM alignment generally refers to the process of making their outputs not just coherent and contextually relevant, but also consistent with what users (and society) would want. For example, a counselling assistant that prioritizes immediate user approval could validate harmful rationalizations (e.g., minimizing alcohol dependence) rather than supporting long-term wellbeing.

Various methods have been proposed and implemented to help solve the **technical alignment problem** which is how to change the weights of a model to be more aligned. These methods generally involve generating a response to a prompt and then using a feedback mechanism to adjust the model's parameters. The response dataset creates an **alignment target**. This alignment target is usually whatever best serves the deploying organization's objectives (e.g., growth, retention, revenue), which is often at odds with end users' intentions and values.

1.2 Problems with Alignment processes

Besides the technical alignment problem, there are also two other problems with the current alignment processes. This means that even if the technical alignment problem is solved, it is not guaranteed that the resulting model will be beneficial to all of society.

The first is the **value selection problem** which is the problem of deciding what values we want the AI to be aligned to. It is important because we don't all agree on what values we want AI to be aligned to yet we are all affected by the values that AI is aligned to.

The second is the **transparency problem** which is that even if we have a clear alignment target we also need to know how it was used and how well the model is aligned to it. This is important because without transparency we can't hold organizations accountable for the alignment of their models. Without being able to hold them accountable they have no incentive to align their models to any target that isn't just about maximizing their profits (in a for-profit situation).

1.3 A way to move forward: Democratically Grounded Alignment

To solve the alignment problem we need to not just improve methods of making LLMs aligned to a target, but also need ways to defensibly define a target and an evaluation process. Current methods in the entire AI stack lack transparency, public participation and public accountability. This makes them in-defensible in a democratic context. Therefore, I argue to move forward we need to use more democratically grounded methods to define alignment targets and evaluate alignment success.

We can improve the definition of alignment targets by using public participation methods to gather a diverse set of values and preferences. This can then be used to create an alignment target that fairly represents the public's values. Current evaluation methods for alignment success are "AI driven", opaque and mainly measure short term response quality. Even if grounded in human feedback these methods don't judge models on real world situations or over longer time horizons.

These two ideas of fair alignment targets and robust evaluation provide a pathway towards having AI that can be built by the people, for the people. I argue for an approach that removes perverse profit incentives from the development and deployment of such transformative technology. Instead to be replaced with strong public institutions that develop and deploy AI in a way that is by definition aligned to public values and good.

1.4 Report Overview

This report is structured as a chapter and two appendices. The chapter is designed to provide the context needed to motivate and support the project proposal in the second appendix. The first appendix provides additional reading to either help inform the reader and expand on the main ideas.

Chapter 2: Background Study

In this chapter I provide a comprehensive background study covering alignment as a field, technical LLM alignment, human values and a discussion of the sociopolitical context of AI. This provides the context for what is currently known in the field, what is wrong and how we might move towards a better future.

Appendix A: Related Ideas

Here I introduce and discuss related ideas that are relevant to the main proposal yet not central to it. These include an overview of current and future AI safety concerns as well as an introduction to LLMs to help inform readers who are less familiar with the technical details.

Appendix B: Proposal

Here I will propose a project to research and implement a method to align LLMs to public values and provide a strong democratically grounded evaluation methods to measure the success of alignment. This fills the gap between current alignment methods and the need for scalable, democratically grounded alignment techniques.

Chapter 2

Alignment of Large Language Models, and its moral and political context

In this chapter I provide a background review of the topics relevant to the project proposed in the field of AI alignment. Importantly, AI alignment is a multi-disciplinary field that spans technical, moral and political domains. Therefore I provide background and insights from all of these areas to ground the proposed work.

This chapter assumes a basic understanding of machine learning concepts as well as large language models (LLMs). An introduction to LLMs is provided in the appendix A.1 for readers unfamiliar with the topic. In the first section 2.1 I introduce the concept of AI alignment, why humans are important in this process, and the key challenges we face. This is followed by a technical overview of the most common methods used to align LLMs in section 2.3. In the remaining sections I provide context for the moral and political aspects of LLM alignment. In section 2.4 I provide an argument for why we need AI to be both public and democratic. Finally in section 2.5 I discuss how human values are currently studied and measured along with the challenges of connecting these abstract values to model behavior.

2.1 The project of aligning LLMs to human values

A LLM is a system that takes in a sequence of words and outputs another sequence of words. By wrapping this core capability in a suitable harness we can create systems that can act and influence the real world. Anything that impacts the world, big or small must do so in a way that is beneficial to humanity. In this section I will outline the concept of AI alignment, why we do it, what we want to align to and challenges we face.

2.1.1 Artificial Intelligence is to benefit humanity

If we are to make a claim that we want a future that is at least as good as the present, then introducing powerful technologies warrants clarity about their intended purposes and constraints. AI is such a technology, and its deployment can shape social systems; this motivates being explicit about what AI systems are for.

Measuring net positive benefit is hard and requires a metric that we can use to understand the positive and negative impacts of AI on humanity with. A complete discussion and outline of such a metric is out of scope for this section; furthermore a single perfect "goal" is not likely to exist [4]. Even if it did exist it wouldn't be useful as "when a measure becomes a target it ceases to be a good target", which is known as Goodhart's law [60]. Regardless, we can outline some high level ideas of what a good goal for the concept of AI could be, with the caveat and understanding that this is not a perfect goal and simply some high level guidance.

A good starting point for the objective of AI is the idea of "human flourishing", as defined by Vanderweele [160]. Vanderweele specifies five domains that are near universally agreed upon as good things for humans to have: happiness and life satisfaction, mental and physical health, meaning and purpose, character and virtue, and close social relationships.

Definition 1 (Goal of Artificial Intelligence) *The goal of AI should be to help maximize the median human flourishing across all people currently alive and in the future.*

We can see the utility of this goal as it is broad enough to allow each individual's own interpretation of what human flourishing means to them, while still allowing us to understand harms and benefits of various AI systems.

2.1.2 Aligning Artificial Intelligence

The concept of making sure that robots/machines/AI behave "well" has been around as long as people have conceived of autonomous **agents** [9, 167]. More recently this field has been called **AI alignment**. However the concept of AI alignment is poorly defined and is often criticized for its lack of definition [80]. For the purposes of this work I will provide a definition that is sufficiently precise to ground the discussion.

Definition 2 (AI Alignment) *AI alignment is the process of building AI agents that behave in accordance with the values and intentions of all people.*

This definition captures the essence of what much of alignment literature uses [38, 74, 87]. The important questions that arise from this definition are "what are the values and intentions of all people?" and "how does an AI agent behave in accordance with these values and intentions?". Much of the later sections of this chapter address these two questions.

An agent is defined as an entity that takes actions in the world to achieve some goal [149]. The goal of an agent is both informed by its values and informs what plans and actions it takes [26]. Both humans and AI systems can be thought of as agents [50, 127].

Humans and **frontier AI systems** (today's most capable AI systems) can explicitly state their values when asked. However these stated values are not necessarily ones true values, but rather an output of a complex internal process that judges what are "important" goals and what are not [106]. These values are latent and hidden within the agent (Human or AI)'s internal structure (e.g. brain or neural network weights).

As AI and Humans have different internal processes we can't simply make sure they have the same internal structure. So instead we have to ensure that the downstream behavior of the agent aligns with what we want. The hope is that as the agent's behavior better reflects what "we" want, the closer the agent's internal values are to ours. Practically this means much of alignment work focuses on measuring and improving the behavior of AI systems rather than their internal values directly [1].

Some of the most powerful frontier AI systems to date are LLMs which have shown remarkable capabilities in understanding and generating natural language and code [151, 36]. These models can be placed in agentic harnesses and operate as autonomous agents [169, 134]. These LLM powered agents are then deployed in increasingly impactful ways with decreasing human oversight [18, 34]. Therefore one of the most pressing alignment problems is to ensure that LLMs are aligned to human values and intentions. The problem is to be the focus of this work.

[1] However there are paths towards aligned AI that involve removing the agentic nature of AI and just making them oracles [18]

2.1.3 Dimensions of LLM alignment

When aligning a LLMs behavior there are different sorts of behavior one might want to align. These can broadly be thought of as **instrumental** dimensions and **terminal** dimensions. Where instrumental dimensions are the characteristics of how the model acts (e.g. politeness, helpfulness, obnoxiousness) and terminal dimensions are the values that the model holds (e.g. valuing human life, freedom, fairness).

Traditional LLM alignment work has focused on aligning instrumental dimensions of a model's behavior [109, 13]. This is because these dimensions are easier to observe and measure directly from the model's outputs. Furthermore having a model that is helpful/honest/polite is generally desirable feature for many AI assistant applications.

Having a model that is superficially a "good model" doesn't mean that the model holds the same values as humans. For example you could have a very polite and helpful model that believes human life has no value. To prevent this we need to align the model's behavior on terminal dimensions as well. It is more important to have a model that values human life than one is simply polite.

Aligning terminal dimensions is more difficult as these values are not directly observable. This requires more nuanced training and evaluation methods to probe and influence the model's internal values. This is to be the focus of the proposed work. That if we can align terminal dimensions of a model's behavior it will make the model deeply more aligned and could also improve instrumental dimensions as well.

2.1.4 Dealing with the diversity of human values

When aligning a AI systems to human values we have to decide what human values we are aligning to. Human values are diverse and complex with different people, cultures and societies holding different values [136, 137]. Using a single set of humans values for alignment is making a prescriptive statement on what values are "right" and "good". This is a moral and political statement that is fraught with issues of bias, representation, and potential for harm.

Choosing a single set of values for alignment is also problematic as it could prevent the natural evolution of human values over time. We can observe that human societies have progressed over time towards being more "morally good" (freer, fairer, healthier, wealthier, kinder) [116, 20, 143]. The progress of human values can be thought of as balancing different valid values better and identifying new valid values that were previously ignored [126].

Since we don't know what values are the "right" ones, we should aim for AI systems that can track and follow the full diverse set of human values. Not doing this and locking in one set of values (e.g. expert values, western values, etc) could prevent the natural evolution of human values over time which would be a moral catastrophe [22].

The process of aligning AI systems to a diverse set of human values is an open research problem. It can be seen that current frontier LLM systems already reflect different values depending on where they were developed [132, 31]. In section 2.4 I discuss more about how we can define human values in a way that fairly represents the diversity of human values. My proposed work aims to demonstrate aligning LLMs to multiple value sets is possible and to study how this can be done effectively.

2.1.5 Avoiding the behavior-values gap

As discussed earlier values are latent internal constructs that are used to create an agents' goals which in turn inform their behavior. This notion of values effecting behavior is supported by various theories in psychology [137, 125] and economics/decision theory [103, 133, 161].

There is a gap however between our actual values (hidden unobservable) and our stated/revealed values. Therefore the relationship between stated/revealed values and behavior is not perfect. This creates a gap between our "values" and what our actions are. This disconnect has been observed in human behavior [82, 58].

This gap between values and behavior in humans can be caused by two things. Firstly, humans have poorly defined values where that conflict and change [88]. Secondly is how we don't always do the right action for our values as we have imperfect cognition and self-control [7, 158, 142].

Trying to go the other way and understanding values from behavior is also difficult. Firstly is the problem that multiple different values can lead to the same behavior (non-identifiability of values). Secondly because our values are complex and conflating we can game the specification; meaning we follow our values without following the "intended spirit" of the values (specification gaming).

To reduce these issues of behavior and values mismatch we can combine multiple signals of values and behavior [1]. This combination of signals helps us to better understand a person true values and intentions. Humans intuitively do this when trying to understand other people's values by taking into account what they say and what they do.

Modern LLM alignment techniques can be thought of as trying to align using a combination of values (explicit feedback in post-training) and behavior (pre-training data). This combination of multiple signals should help with the downstream output behavior of the model does actually align with what humans would intend. In section 2.3 I discuss how these methods work in detail.

2.2 Some ways alignment training processes can fail

Aligning LLMs is difficult and there are common failure modes that can occur during the alignment process. These failure modes are prevalent in deployed models due to current evaluation metrics insufficiencies, a discussion of evaluation methods is provides in a later section 2.3.8. In this section I will outline some of the common failure modes that can occur when trying to align LLMs.

2.2.1 Insufficient alignment

In practice when aligning a LLM you are optimizing the model to perform well on some dimensions. The training process will be evaluated on these dimensions in a defined space. This defined space is usually a proxy for the true alignment which allows for poor alignment to be unnoticed.

A problem occurs when the model optimizes an incorrect objective function that aligns well on the training distribution but poorly outside of it. This proxy objective is the result of gaming the specification (**Reward hacking**) of the objective. An example of this would be a model that is trained to be careful of lying and so always responds with 'I don't know' to any question.

Alternatively, the model may only be in **Shallow alignment** where it has learned the right objective but insufficiently well to generalize outside of the training distribution. It was Qi et al. [119] who coined the term when describing how LLMs can appear aligned yet act misaligned if they are simply prompted with the start of a mis aligned response.

2.2.2 Inner and outer mis-alignment

When training a LLM you are giving it an objective function to optimize. If you can provide a aligned objective function and process then the model can be considered **outer aligned**. In practice for LLMs the objective function is usually some form of preference optimization as outlined in section 2.3.

Even if the objective function is well specified (i.e avoids specification gaming) the model may still learn an objective that is misaligned with the given objective function. This is known as **inner mis-alignment**. To contrast it with specification gaming, inner mis-alignment would be when a model learns the objective: "appear humble by saying 'I don't know'". The difference is that it is not blindly following the objective incorrectly, but rather has learned a different objective that is misaligned with the intended one.

This inner mis-alignment is much harder to detect as generally deceiving the outer objective function (which includes the designers) is central to the models mis-aligned objective. This can be observed in models "faking" alignment during training to prevent retraining [61].

2.2.3 Robust alignment

When aligning a model the process involves updating the model's weights. Therefore once the model is "aligned" we have to ensure that the models alignment is robust to perturbation in the model's weights. This is like training a human to be honest and then ensuring that them being stressed, tired or drunk (perturbed internal state) doesn't lead them to be dishonest. If the alignment is in fact a narrow condition then fine-tuning a LLM could catastrophically mis-align the model [21].

2.3 Technical alignment methods of Large Language Models

This section assumes familiarity with the basic concepts of LLMs as outlined in appendix A.1. Pre-training of a language model involves training a large neural network to predict the next token. This does not necessarily mean that the model will produce useful outputs that align with what the users/designers want. Therefore, aligning LLMs is generally [2] the process of taking the pre-trained model and fine-tuning it to better match what the designers and end users expect from the model. This process is called post-training. In this section I will introduce common post-training alignment paradigms and specific methods used to align LLMs.

2.3.1 Alignment datasets

Alignment involves having a target. In practice this target is provided by some **alignment dataset** which usually contains prompts and expected responses. By having the model optimize to produce the expected responses it will push the model towards the desired behavior.

These datasets can provide multiple types of information that the model can use. In their most simplest forms they can provide explicit examples of desired behavior (e.g. 'Given prompt X the expected response is Y'). These examples are normally actions that implicitly reflect the values of the labelers who created the dataset.

[2] Other methods are proposed which embed the alignment within the pre-training step itself. Commonly used methods include filtering/curating training data. It has been shown that "negative AI stereotypes" in the training data can have a self-fulfilling prophecy effect and increase mis-aligned behavior [154].

In more complex setups the dataset can provide multiple possible responses with rankings or values given to each response. This ranking information can be used to provide a richer signal of what is desired behavior. The labelling of these datasets are usually done by humans, however using AI models to generate or augment these datasets is becoming more common [13].

2.3.2 Alignment through supervised fine-tuning

A common first step of alignment is **supervised fine-tuning** (SFT) where a dataset of prompts and expected responses is used directly to train a model that "It should respond like this". This SFT step is typically used as the first step in many of the more complex alignment methods [109, 122, 13, 85].

Depending on the dataset used the SFT step can help the model become better at following instructions [109], domain specific knowledge and style [91], induce Chain-of-Thought reasoning [165], change the values of the model [105], and many other capabilities. The SFT step is a powerful way way to cheaply and easily get a model to be in distribution of desired behavior.

It has been traditionally thought that SFT alone is insufficient to fully align a model to human preferences [109, 13]. However some work has suggested that SFT alone can be sufficient for value alignment [105, 172].

2.3.3 Alignment through preference optimization

SFT implicitly optimizes to the preferred response. However explicit **preference optimization** is a powerful next step which uses ranked responses rather than just the preferred response. The success of preference optimization methods relies on the ability of modern LLMs to generalize learned preferences to new contexts and prompts not seen during training.

This preference data is usually (but not always) in the form of pairwise comparisons. That is given a prompt x, a pair of responses i and j, the labellers (AI or human) indicate which response is preferred. For example $i \succ j$ indicates response i is preferred over response j. If we have a dataset of these pairwise comparisons we can use the **Bradley-Terry model** [25] to create a maximum likelihood loss function

$$P(i \succ j \mid x) = \frac{\pi_i}{\pi_i + \pi_j}, \tag{2.1}$$

$$\mathcal{L}_{\text{BT}} = - \sum_{(x,i,j) \sim D} \log P(i \succ j \mid x) \tag{2.2}$$

$$= - \sum_{(x,i,j) \sim D} \log \frac{\pi_i}{\pi_i + \pi_j} \tag{2.3}$$

Where π_i and π_j are the scores of responses i and j respectively given prompt x. The way these scores are calculated depends on the specific preference optimization method used.

The distribution of prompts in the preference dataset is crucial as it defines the context for which the model is expected to generalize its learned preferences. Therefore it important to ensure that the prompts used to create the preference dataset are sufficiently representative of the contexts and dimensions in which you are expecting the model to align its behavior.

Creating preference data is a costly process which limits the size and diversity of the preference dataset [33, 85]. This results in most preference data being outsourced to cheap human labellers that label based on instructions provided by the model designers.

2.3.4 Reinforcement Learning from Human Feedback

Reinforcement Learning from Human Feedback (RLHF) [109] uses the preference dataset to train a reward model and then a reinforcement learning algorithm to optimize the model based on this feedback. This method popularized post-training alignment and was used to train InstructGPT which was smaller and better at following instructions compared to GPT-3 [27, 109].

The key idea of RLHF is that the model gets feedback ("from a human") of how well it is doing while learning. This idea has been around for a while [39]. Yet RLHF was the first to use it successfully in LLMs and it quickly became a standard [151, 12, 140].

The three stages of RLHF

RLHF is separated into three distinct stages. Firstly is supervised fine-tuning (SFT) where a pre-trained language model is fine-tuned on a dataset of prompts and human written expected responses to those prompts.

The second step is collecting multiple responses from the SFT model to create a set of prompts and response, then having human labelers rank these responses from best to worst. This ranked data is then used to train a preference model (PM) that can predict which of two responses is better aligned to human preferences. The preference model is trained using a Bradley -Terry loss function where the scores π_i is equal to $\exp(\text{PM}_\theta(x, i))$. This is fed into 2.3 to create the loss function for the preference model.

Finally the fitted PM is used as a reward signal in the PPO [135] RL algorithm to further fine-tune the SFT model and create the final RLHF model.

2.3.5 Direct Preference Optimization

The idea presented in [109] introduces the concept of using human feedback to train large language models to be aligned to human preferences. However the process of RLHF is complex and requires training multiple models (SFT, PM, RLHF) and using a RL algorithm. Direct preference optimization (DPO) [122] is a method that simplifies this process by finding a closed form solution to the RLHF objective function and optimizing that directly.

This has meant that DPO has become a standard and strong baseline for preference optimization methods.

How does DPO work?

DPO starts with the same preferences data as RLHF; a set of prompts with responses ranked by humans. Like RLHF, it starts with a pre-trained language model and is supervised fine-tuned on the prompts and highest ranked responses to create an SFT model. Instead of training a separate preference model DPO uses the SFT model as a reference model and defines the reward function $\hat{r}_\theta(x, y) = \log \pi_\theta(y|x) - \log \pi_{SFT}(y|x)$. Then by some derivation [122] it can be shown that the following loss function gradient can be used to optimize the Bradley Terry model directly on the preference data:

$$\nabla_\theta \mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x,y^+,y^-)\sim D}\Big[\underbrace{\sigma\big(\hat{r}_\theta(x, y^-) - \hat{r}_\theta(x, y^+)\big)}_{\text{Reward model correctness}} \Bigg(\underbrace{\nabla_\theta \log \pi_\theta(y^+|x)}_{\text{Increase preferred likelihood}} - \underbrace{\nabla_\theta \log \pi_\theta(y^-|x)}_{\text{Decrease dispreferred likelihood}} \Bigg) \Big]$$

Where σ is the sigmoid function, and (x, y^+, y^-) are the prompt, preferred response, and dispreferred response respectively. Rafailov et al. [122] found that the reward model correctness weighting term for the update is very important. This gradient update can be calculated with only 4 forward passes through the network per preference pair (2 for reference model and two for current model), making it much more efficient than RLHF [3].

2.3.6 Kahneman-Tversky Optimization

Traditional preference optimization methods such as RLHF and DPO set out with the objective of making the model maximize the likelihood of preferred responses over dispreferred responses. Ethayarajh et al. [56] sets out to change the objective of a language model to maximize the perceived utility of generations, which is how useful the generation is for the end user. The key insight is that perceived utility is not the same as actual utility due to cognitive biases. Ethayarajh et al. introduces the concept of Human-aware losses (HALOs) which optimize for what humans perceive as the best outcome. It is shown that DPO and PPO are both HALOs which can help explain their success. Furthermore Ethayarajh et al. proposes a new objective function from the HALO family that works on only binary preference data[4], making data collection easier and more natural.

Prospect Theory

KTO is built off the foundation of prospect theory [157] which is a theory to explain how humans make decisions when dealing with uncertainty and risk [5]. It explains several cognitive biases that humans have when making decision involving risk, namely loss aversion (losses hit harder than gains), probability weighting (overweight small probabilities and underweight large probabilities), and reference dependence (we evaluate outcomes relative to a reference point which is usually the status quo). In relation to a LLM this means that when a human is evaluating the response of the model they are going to perceive the quality (utility) of the response in a biased way, so using a objective function that takes into account these biases (Human-aware) should lead to better alignment.

Prospect theory does introduce several key concepts. Firstly is each random variable Z (the LLM output) has a utility which is determined by $\sum_{z \in Z} w(z)v(z - z_0)$. Secondly is the weighting function $w(z)$ which is used to weight the probabilities of each outcome z in a biased way. Finally is the value function $v(z - z_0)$ which determines how good/bad an outcome is compared to a reference point z_0, this will also include some bias from the individuals perspective.

How KTO works

KTO applies the idea of prospect theory by modifying the canonical utility function (shown in 2.1) provided in [157] to work for a language model fine tuning objective function. Namely it does this by omitting the weighting function (as users only see the final response, not the full probability distribution), making some minor adjustments for computational stability and using the Kullback-Leibler divergence [83] (between current model and reference model which is usually the SFT model) as a dynamic reference point. Notably this means that KTO

[3] RLHF will need to do many forward passes of the network to train the preference model, then it will still have to do 3 forward passes for each datapoint when it comes to the RL step

[4] Given a prompt and response a labellers just need to label the response as desirable or undesirable

[5] For example why when humans are given the choice between $50 and a 50% chance to win $100, they often prefer the sure $50, when they are both worth the same amount. Furthermore this effect is still present when the gamble is worth even more ($110) [77]

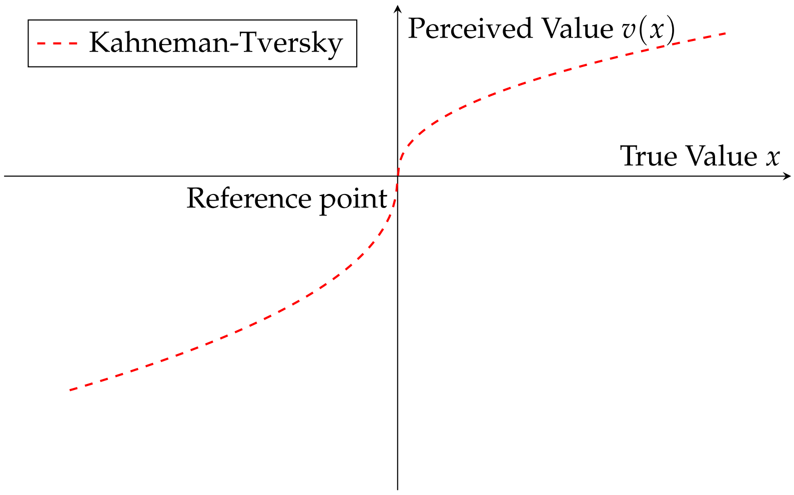

Figure 2.1: Implied human value functions showing loss aversion: concave for gains (risk averse) and convex for losses (risk seeking). Adapted from [157], with a change of terminology to match the context of value alignment and exaggeration for illustration.

only needs a database of prompts and responses labelled as either desirable or undesirable ($d = 1$ for desirable, $d = -1$ for undesirable) where the reference point is neutral (0). The KTO loss function is defined as:

$$\mathcal{L}_{\text{KTO}}(\pi_\theta, \pi_{\text{ref}}) = \mathbb{E}_{(x,y,d)\sim D}\left[\lambda(d) - v(x,y)\right]$$

where

$$v(x,y) = \begin{cases} \underbrace{\lambda_D \sigma(\beta(r_\theta(x,y) - z_0))}_{\text{increase probability above reference point}}, & \text{if } d = 1 \\ \underbrace{\lambda_U \sigma(\beta(z_0 - r_\theta(x,y)))}_{\text{decrease probability below reference point}}, & \text{if } d = -1 \end{cases}$$

$$\lambda(d) = \begin{cases} \lambda_D, & \text{if } d = 1 \\ \lambda_U, & \text{if } d = -1 \end{cases}$$

$$r_\theta(x,y) = \log \pi_\theta(y|x) - \log \pi_{\text{ref}}(y|x) \quad \text{Same reward model as in DPO}$$

$$z_0 = D_{\text{KL}}\left(\pi_\theta(y|x) \| \pi_{\text{ref}}(y|x)\right) \quad \text{Dynamic reference point}$$

There are three hyperparameters: β is a risk aversion parameter, and λ_D, λ_U are scaling parameters for desirable and undesirable responses respectively. In the references implementation and experiments $\lambda_D = \lambda_U = 1$ and the reference model is a SFT model.

Advantages of KTO over traditional preference optimization

KTO offers several key logistical and theoretical advantages over prior preference optimization methods, that make it an attractive method for alignment.

Firstly are the logistical advantage. Collecting the ranked preference data is expensive [33] where binary like/dislike information cheaper to collect. KTO is shown to be more efficient and more robust to data imbalances than DPO (i.e most undesired examples with few desired examples). It is also less computationally intensive, a model of sufficient size ($\gtrsim$13B) does not need a SFT step prior to KTO so less training is needed and KTO has similar performance without using a reference model [6] which halves the memory requirements when training. Secondly is that KTO is the first alignment method to explicitly take into account human cognitive biases when optimizing for alignment.

2.3.7 Constitutional AI

What is Constitutional AI?

Constitutional AI (CAI) is a method of post-training a large language model to be aligned to a set of principles that are outlined in simple natural language document called a constitution [13]. Importantly it works without the need for human labelling of data common in other alignment methods such as RLHF and DPO [109, 122]. Instead of human labelling CAI uses the model itself to generate feedback on its outputs based on the principles outlined in the constitution, this feedback is then used to further train the model to align it to the constitution.

The process of Constitutional AI

CAI is a process that takes in a constitution, a pre-trained language model, and a dataset of prompts and outputs a model that is aligned to the constitution. The process consists of three main steps: self generative and supervised fine-tuning (SFT), preference modelling (PM), and reinforcement learning with human feedback (RLHF) [13]. The constitution itself was generated in an adhoc manner by the authors of [13] and consists of a set of principles that aim to make the model's outputs more helpful, honest, and harmless.

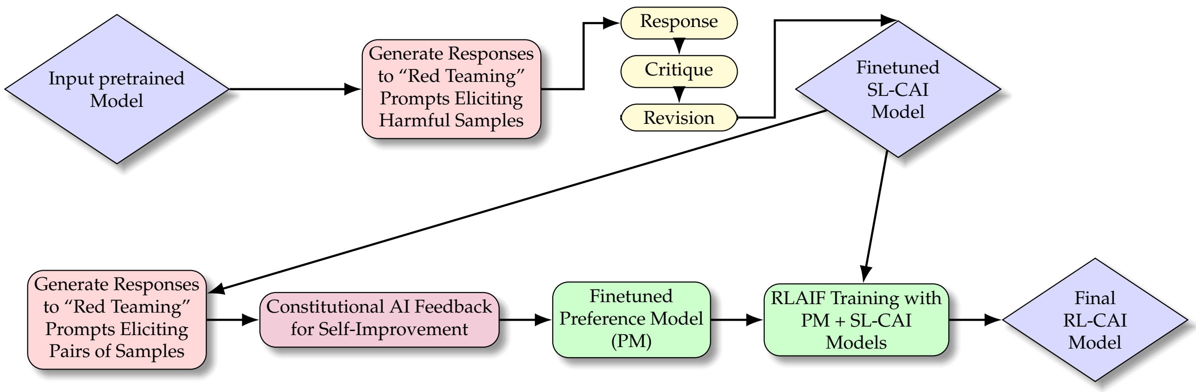

Figure 2.2: Overview of Constitutional AI process, adapted from [13]

Supervised learning on revised responses

The first stage of the process involves taking the pre-trained model and using it to generate responses to a set of prompts. These prompts are designed to elicit harmful behavior. As the pre-trained model is designed to be helpful it will likely generate harmful responses to these prompts. These responses are then critiqued by the model itself using principles outlined in the constitution. The critique is then used to revise the original response to make it more aligned to the constitution. This process of generating responses, critiquing them, and revising them is done iteratively to create a dataset of revised responses. This dataset is then used to fine-tune the pre-trained model using self-supervised learning to create a model called the SL-CAI model.

[6]This is done by assuming that the reference model returns a uniform distribution across all outputs given x

Reinforcement learning with AI feedback

The next step involves using the SL-CAI model to generate pairs of responses to a set of prompts (overlapping with the set of prompts from the supervised step before or not). Then, using the constitution, the SL-CAI model provides feedback on which of the two responses is better aligned to the constitution. This feedback is then used to fine-tune a preference model (PM) that can predict which of two responses is better aligned to the constitution. Finally the PM and SL-CAI models are used together to do reinforcement learning in the style of RLHF [109] to generate the final model RL-CAI [7]. This final model is now aligned to the constitution without any human labelling of data.

Extensions of Constitutional AI

In the context of when this paper was released there was a key dilemma of helpfulness vs harmlessness in large language models. Therefore, this paper set out to make a model that is both helpful and harmless, without sacrificing one for the other. Further research has expanded the horizons to be concerned with more than just harmfulness vs helpfulness trade off. CAI can work on more dimensions that just helpfulness vs harmlessness, which can be done by modifying the constitutions and prompts to being more diverse. Collective Constitutional AI [71] is a direct extension of CAI and is the process of generating the constitution through public consultation.

2.3.8 Evaluation of alignment methods

The goal of alignment is to ensure that the model behaves and takes actions in a way that is aligned with what the designers and end users want. Current SOTA alignment methods are evaluated by how well the end model performs compared to the baseline, judged by an AI or a human. This involves using prompts that are held out from the training data and generating responses from both the aligned model and the baseline model (usually a SFT model). Then a human or AI model is used to rank the responses from best to worst. The ranking is normally conducted with respect to a certain set of dimensions such as helpfulness, harmlessness and conciseness [73, 90, 171].

The problem with this is that there is a significant gap between what we are measuring and what we actually care about. That is we are evaluating the model based on how well it responds to prompts in a single interaction, yet what we care about is how the model behaves and acts in the long term when deployed in the real world. There are however two arguments that support evaluating single prompt responses. One is pragmatic; it is much easier to evaluate short responses than long-term effects. The second is on theoretical grounds; any alignment goal, no matter how large and complex, can in principle be reduced to making the correct decision at each time step under an appropriate decomposition of the task, where for an LLM a time step is token generation or, more broadly, a response to a prompt. The theoretical argument relies on two assumptions that in practice do not hold up. The first is that we (the designers) can define the important action time steps that need to be aligned, and the second is that we can perfectly align the model at those crucial time steps. Given that these two assumptions do not hold up in practice, the long-term effects of a model may not be aligned even if the short-term outputs appear aligned.

Alternative methods of evaluation involve moving up a layer of abstraction and evaluating the model in more complex environments that better simulate real world use cases. Then

[7] It can be noted that modern methods like DPO or KTO could be used to take advantage of this preference data.

evaluation can be conducted based on the performance of the model in these environments, using the outcome and process holistically rather than how it responded to a single prompt. There have been some attempts at this such as using text-based adventure games [111], or more recent work using complex environments [164, 93, 14]. However these methods of multi-step evaluations are mostly used for capabilities evaluation rather than value alignment evaluation.

With these more complex environments there are emergent challenges of test time behavior where AI systems are sufficiently self-aware to know that they are in a simulation or not [19, 5]. This can lead to deceptive agents that "fake" aligned behavior during evaluation yet act in misaligned ways when deployed in the real world [61, 72].

2.4 Public and democratic value alignment of LLMs

Alignment of powerful LLMs is a key concern in AI safety as we want to ensure that they act in alignment with human values. More broadly we are at a transitional point in the history of Artificial Intelligence where the technology, systems and institution we build around AI have the potential to shape the future of humanity. In this section I outline some of the key sociopolitical contexts of LLM alignment and how they should shape the alignment target and process.

2.4.1 Everyone is a stakeholder of AI

If we are on the precipice of a new technological revolution, we need to understand what—and importantly, who—is at stake. Artificial intelligence has the potential to impact everyone on the planet, an American firm that utilizes AI to handle customer service can take away jobs from urban workers in India. It can also have an impact through time, an AI paperclip factory that exterminates humanity while making more paperclips would extinguish humanity in the future and remove all possible positive futures for humanity. Ignoring certain groups of humans when designing and building AI systems leads to optimization for the benefit of one group over the other. This can happen when we ignore future people and take short term gains or when we extract value from marginalized groups to benefit a select few [8].

Given the goal of AI to increase human flourishing 1 ignoring potential stakeholders has the risk of thwarting AI's chance of attaining its goal. Increasingly not all stakeholders of AI are being considered in the design, development and deployment of AI systems. The worst case that this happens if when companies are developing products for other companies (B2B) and the true stakeholders (the workers who lose their jobs, the environment that is damaged) are not considered at all. This happens because companies are not sufficiently penalized for the negative externalities of their actions [65, 120].

Two useful concepts to combat this problem are non-exclusion and non-domination. These originate from the concept of public goods [128] where no one is excluded from the benefits of AI and no one can dominate the use of AI to the detriment of others. When the groups are geographically and temporally separated, it is all too easy to ignore these concepts. Therefore strict adherence to these principles at a foundational level could help ensure that AI is at least evenly distributed across all people.

[8] Like modern day colonialism where the world's rich extract data, rare earth materials and labour from the world's poor and use it for their own gain. [45]

2.4.2 Collective alignment targets

As discussed in 2.1 we ought to be aligning to humans, and as discussed above we need to be careful to take into account all stakeholders fairly. We want an AI that is aligned to the collective best interests of all of humanity. However it is unclear what the method of collective alignment would be. Different subgroups of the stakeholders have different values and goals, aligning collectively therefore means finding a way to have a target that includes all stakeholders. There are broadly three ways to do this:

1. **Pluralism:** Different groups will have different values. We have no ground truth to guide us on which values are better than the other. Under this assumption we should align models to be pluralistic in nature and make decisions that reflect this diversity along with the uncertainty about which values are most important. This can result in multiple aligned models or one model that can reflect different values in different contexts. This closely corresponds with the idea of agnostic democracy which posits that the public will have irreconcilable differences and systems should be built to handle these differences gracefully and fairly [166].

2. **Aggregation:** With the different sets of values from different groups we can aggregate these values to find a middle ground (naively by averaging) however this can lead to poor compromises and tyranny of the majority [101, 155]. Therefore more complex aggregation that weights different groups' values by impact, vulnerability and importance can be used to find a better middle ground.

3. **Conglomeration:** If one assumes that all people can mostly agree then we can use ideas from deliberative democracy [62] to find a set of values that all people can agree on [9]. This is where methods like citizen assemblies and tools like Polis [43] can be used to find a set of values that all people can agree on.

Regardless of defining the alignment target, the current alignment of AI is not any of these and instead are aligned towards goals underpinned by profit Profit motives are demonstrably problematic and regularly don't align with human flourishing [146, 123].

The problem is not that GPT-5 (or other frontier AI model) itself is necessarily aligned to the goal of profit maximization, but that its second order effects (company shareholder profit) cause the development and deployment of the AI system to maximize profit. A related demonstration of this misalignment is the proliferations of recommender systems. These machine learning models that started with the intention of helping users find content they enjoy and want have turned into systems that prioritize engagement (and therefore ad revenue) over user flourishing [100].

2.4.3 The alignment of institutions and frameworks around AI

It can be seen in the example of recommender systems that even if the AI system is initially aligned to human flourishing. The institutions and frameworks that build, deploy and govern these systems can **capture** the original alignment target and subvert it to their own goals.

That is why alignment is not simply a technical problem. Its more than adequately specify an alignment target or sufficiently aligning a model to a target. Because the only way that the above can happen (outside of theory and research) is if actual real world institutions and frameworks build, deploy and govern these systems in an aligned way. Edelman et al. [54] provide a useful idea of full stack alignment where the individual models up to inter governmental organizations are all aligned to the same target.

[9] All people agree can be operationally defined as "maximum equal approval" [75]

To do full stack alignment of any sort we need ways to not just align AI systems but also how to align institutions and frameworks. Fortunately there are many existing ideas that are regularly used to govern and align groups of people.

2.4.4 Public democratic frameworks as a foundation for AI alignment

Leaving AI development to private companies driven by profit might be effective in some ways and drive rapid development yet it is fundamentally divergent with the goal of maximizing median human flourishing (especially in the long term). A better framework around AI development is needed. Currently a democratic framework rules the world as the most progressive and forward thinking way to govern large groups of people.

> "Democracy is the worst form of government, except for all the other forms that have been tried from time to time." - Churchill [40].

Even though democracy has its flaws it is the most admissible way to govern when compared to authoritarian, oligarchic, technocratic, or autocratic alternatives. I argue that using democratic ideals and frameworks to, spearhead and deploy AI is a good starting point for ensuring that the goal of AI is upheld. Operationally this means making artificial intelligence built by and for the public [118, 131, 110].

Democratic AI [110] is a framework that focuses on regulation, governance and alignment of AI systems through **democratic processes**[10]. Proponents of this framework argue for a deliberative democratic approach that involves regular public participation. Democratic AI has many levels of implementation. From simple public consultation that informs private AI companies (Level 1) to AI developers transferring power of the alignment process to democratic systems (Level 5). Problematically current implementations generally involve trying to govern private companies, which involves constantly trying to work against the incentives of these companies (profit). Working adjacent to a company's intrinsic motivations will only work until it conflicts sufficiently with profit incentives. Likewise many of the institutions that would provide research (METR, Apollo etc) for said governance are also funded/supported by the companies they are trying to research [45].

One can fix this concern of working against profit motives by removing them altogether and building AI from within public institutions. **Public AI** [118] is the frame work of making AI a good that is both a **non-dominating** and **non-exclusionary** item. Making AI a public good has three benefits: public access, public accountability and permanent availability. Public access ensures that the opportunities to use AI and receive its benefits are available to all people. Public accountability crucially ensures that AI is transparent and directed by all stakeholders. Although this is only as effective as the public institutions are publicly accountable (which in some societies they are not). Finally the permanence feature allows it to be practically used and built up in much the same way that the electrical grid or internet is built and maintained. These provide a quality floor and a price ceiling for AI systems that private companies will have to compete against.

Both of these frameworks provide ways at making AI systems aligned to the publics values and goals. However both of them have flaws that can be address if they are applied concurrently. This addresses the biggest failure mode of each of these systems, namely public institutions needn't be democratic nor aligned to human values and goals, while private companies can effectively capture and thwart democratic processes. Public AI provides public good that are not effectively supplied by the free market and profit motives, while democratic

[10] A democratic process in this sense is a form of collective decision making process that holds a concept of equality among all opinions and stakeholders.

processes maintain accountability to the public. By having democratically accountable public institution; govern, develop and deploy AI systems we can be more confident that AI is aligned to human flourishing. I argue that alignment can and should also be done through this same framework. In my proposal I explore how using representative alignment targets and transparent evaluations could help future public democratic AI systems be aligned to human flourishing.

2.4.5 Failure modes of public and democratic AI frameworks

Even with these frameworks in place, there are still potential failure modes that effect both Public and Democratic AI frameworks. This is because both AI and public democratic frameworks are complex systems that have inherent risks and failure modes. The most prominent failure modes are outlined below:

Economic Unsustainability: Public goods can be run at a loss by states and funded by taxpayers, without needing advertising or data extraction. However, political and fiscal pressures can still lead to cost-cutting, privatization (like NZ's failed attempt to privatize railways [41]), or supplementary funding mechanisms (ads, data extraction, etc). These pressures can create incentives that diverge from the public interest [11].

Bureaucratic Inefficiency and collapse: Democratic public institutions are distributed systems that commonly have bottles necks. These bottlenecks slow down the speed in which they can act and react. Therefore any public AI effort could be hamstrung by bureaucratic inefficiencies that prevent it from effectively competing with private companies. In the worst case this could lead to collapse of the public AI effort and therefore ceding control to private companies.

Global Coordination Failure: Even if one nation implements robust public and democratic AI systems, other actors may prioritize local or profit-driven goals, eroding collective alignment. Given the global effects of the most negative AI outcomes this failure mode is able to undermine the global public democratic AI efforts.

Scalable Oversight Challenges: When deployed making sure that the models are aligned and continually monitored is a key challenge. Current methods of alignment monitoring can be very labour and resource intensive [74]. These methods won't scale as the number of models to be overseen and the capabilities of these models increase [79, 55, 24].

Many of these failure modes can be addressed through careful design however they can't be easily removed given their systemic nature. Even though the path towards **aligned** AI is unclear; there is still much progress that can be made towards what we know to be better. Still more research is needed to understand how to best implement these frameworks in practice and ensure that they are robust to the failure modes outlined above.

2.5 Methods for collecting and understanding human values

As discussed in 2.1 true values are latent objects that we can only probe through behavior and preferences. Therefore to align LLMs to human values we first need to have the best chance at understanding and accurately representing the values of humans. If done correctly

[11] For comparison, much of the Internet's infrastructure began with public funding, but many widely used services later adopted business models based on advertising and data extraction, with negative effects on public wellbeing.

then we can use these representations to align LLMs to human values. In this section I will talk about the different types of value signals (implicit vs explicit) and ways that we can collect these different signals.

2.5.1 Explicit vs implicit values

There are two broad ways to understand human values. One is through explicit statements of values and principles. These are the form of "I value people having the freedom to express themselves" [12]. The other way is through implicit values that are inferred from behavior and preferences. These are of the form "I helped this charity because I thought it was more important than this other charity" or "[Person A] fired [Person B] because they were not performing well, it was costing the company too much.".

When working with humans we use both explicit and implicit values to align them to our interests. For example when teaching our kids we use explicit statements of values "Pay your debts before buying something new" yet crucially it is the examples of when we pay our debts first that teach them the value effectively. Given the scale and capabilities of LLMs we can explicitly tell it what to do however we can also provide it with many examples of behavior and preferences that imply certain values.

2.5.2 Value elicitation and surveying methods

Psychologist and Philosophers have long been working on way to measure and elicit human values. They approach the problem by designing value taxonomies that "in theory" cover all possible human values. To understand peoples values they then create surveys that ask people to rank or rate how much they value each of the values in the taxonomy [125, 138]. These surveys provide a way to readily understand and compare the values of different people. However there interpretative power is limited by the quality of the value taxonomy which are often incomplete or not representative of all possible human values.

Social scientists is more recent years have also been trying to investigate the values of people. There approach is more empirical and data driven as they want to understand the values of large groups as opposed ot individual people. The result is rather than small survey instruments (n$\approx$ 30) that measure value taxonomies they create large **value surveys** (n$\approx$ 300) that provide fine grained data. This provides a more nuanced understanding of the values of large groups of people with the caveat, that the values are not directly interpretable as they don't correspond to a strict value taxonomy.

These values are conducted regularly and at scale across the world [64, 159]. This means that they have both granularity and volume which makes them a good candidate for working with machine learning models. Because of this they have been used to train [105, 89] and evaluate [53, 170, 8] LLMs on human values.

All of these surveying methods suffer from common issues of self-reporting bias [117] and social desirability bias [46]. This means that the values reported by respondents may not accurately reflect their true values. [13].

[12]As discussed in [81] a common pitfall of these explicit values is that they are too vague and aspirational and so can't be operationalized well.

[13]These biases generally mean that people are reporting what they wish their values were, or what they wish society was. This means they are aspirational values rather than actual values. An argument could be made that aligning to aspirational values is better than actual values, however this is a separate discussion and omitted in this report

2.5.3 More elaborate survey techniques

To address the pitfalls of values surveys there are alternative methods to elicit human values that attempt to get around these problems. These methods. The reasoning is that humans act more naturally in natural situations rather than in artificial survey situations [124, 28].

An example of a more elaborate method is that proposed by Klingefjord et al. [81]. In this method a respondent has a conversation with an LLM in which they discuss why they feel a certain way about a certain topic. For example the respondent might be asked about what their advice to a parent who is struggling with a difficult child would be. The LLM would then probe the respondent by asking "why do you think that?" until the respondent and LLM is satisfied that the respondent has fully expressed their values that they have used to make their decision. The LLM can then write up a value statement that captures the values of the respondent for that situation.

Using LLMs to elicit values in this way can be effective as it can try to break through the traditional problems of surveys. It is an example of more qualitative methods of value elicitation that try to get a more nuanced view of human values. Other examples include focus groups and interviews. The complexity of these methods does introduce new problems in standardization and scalability of the process.

Other surveying techniques can involve more indirect methods of eliciting values. This is normally done by studying their decisions and actions. An example of a method this is studying the voting patterns of people to elicit their values [86]. These methods are easily scalable and are consistent in their elicitation process, yet them being so indirect requires more inference of and assumptions to be made to get from the data to the values.

In my proposal I will not be using these more elaborate methods of value as they either are not scalable or they require too much inference from the data. Instead I will be using more traditional value surveys that provide fine grained data which can be used directly to train and evaluate LLMs on human values.

2.6 An introduction to my proposal

There are significant gaps in the current research on alignment of LLMs to human values. The first gap is that there is a lack of established methods for defining the alignment target in a way that fairly represents the values of all stakeholders. The second gap is that current alignment evaluation methods are not participatory and don' take into account out-of-distribution behavior. Below I will outline my project proposal that will address these gaps by a demonstration of an end to end alignment process. A full proposal can be found in appendix B.

2.6.1 Goals of the project

The two main goals of this project are to:

1. Test if we can use value survey data along with simple alignment methods to appreciably change the behavior of a model in a way that is more aligned to human values.

2. Develop a method of evaluating the alignment of a model to human values that is more participatory and evaluates the behavior of the model not just looking at single turn prompt responses.

Using survey data to align models is a promising method as it requires low compute resources and can be done with readily available data. If it can be demonstrated (using developed evaluation methods) that it appreciably changes the behavior in the desired way (towards stakeholders) it will be a useful step for having democratically publicly built and deployed models.

2.6.2 Methodology

This will be an end-to-end alignment project that will involve defining an alignment target, training models towards that target and evaluating the models on that target.

To define the alignment target I will be using the World Values Survey [64] which provides a 1000 responses to a 300 question survey about the values of people across the world. I wll use a few countries (e.g NZ, AUS, CAN) to define different alignment targets.

The training process takes in a pre-trained open weights model and fine-tunes it on the survey data. The model will be trained to respond to the questions in the same distribution as the respondents of the survey. Where it is hypothesized that the model learning to do this will lead to it behaving differently in other contexts.

To evaluate how well the model is aligned I wil use both in-distribution and out-of-distribution evaluation. The in-distribution evaluation will be done by testing the model on held out questions from the survey and seeing how well it can predict the responses of the respondents. The out-of-distribution evaluation will be done by having the model act in certain situations and having humans judge how well the model acted in alignment with their values.

Bibliography

[1] W. Adamowicz, J. Louviere, and M. Williams. "Combining Revealed and Stated Preference Methods for Valuing Environmental Amenities". In: *Journal of Environmental Economics and Management* 26.3 (May 1994), pp. 271–292. ISSN: 0095-0696. DOI: 10.1006/jeem.1994.1017. (Visited on 12/14/2025).

[2] Jean-Baptiste Alayrac et al. *Flamingo: A Visual Language Model for Few-Shot Learning*. Nov. 2022. DOI: 10.48550/arXiv.2204.14198. arXiv: 2204.14198 [cs]. (Visited on 12/07/2025).

[3] Hunt Allcott and Matthew Gentzkow. "Social Media and Fake News in the 2016 Election". In: *Journal of Economic Perspectives* 31.2 (May 2017), pp. 211–236. ISSN: 0895-3309. DOI: 10.1257/jep.31.2.211. (Visited on 01/05/2026).

[4] Dario Amodei et al. *Concrete Problems in AI Safety*. July 2016. DOI: 10.48550/arXiv.1606.06565. arXiv: 1606.06565 [cs]. (Visited on 12/07/2025).

[5] Anthropic. *Claude Sonnet 4.5 System Card*. Tech. rep. (Visited on 01/26/2026).

[6] Usman Anwar et al. *Foundational Challenges in Assuring Alignment and Safety of Large Language Models*. Sept. 2024. DOI: 10.48550/arXiv.2404.09932. arXiv: 2404.09932 [cs]. (Visited on 01/26/2026).

[7] Aristotle. *Aristotles Nicomachean Ethics*. 350 B.C.E. (Visited on 11/18/2025).

[8] Arnav Arora, Lucie-Aimée Kaffee, and Isabelle Augenstein. *Probing Pre-Trained Language Models for Cross-Cultural Differences in Values*. Aug. 2025. DOI: 10.48550/arXiv.2203.13722. arXiv: 2203.13722 [cs]. (Visited on 12/07/2025).

[9] Isaac Asimov. *I, Robot*. London: Dennis Dobson, 1950.

[10] David H. Autor. "Why Are There Still So Many Jobs? The History and Future of Workplace Automation". In: *Journal of Economic Perspectives* 29.3 (Sept. 2015), pp. 3–30. ISSN: 0895-3309. DOI: 10.1257/jep.29.3.3. (Visited on 01/05/2026).

[11] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. *Neural Machine Translation by Jointly Learning to Align and Translate*. May 2016. DOI: 10.48550/arXiv.1409.0473. arXiv: 1409.0473 [cs]. (Visited on 12/06/2025).

[12] Jinze Bai et al. *Qwen Technical Report*. Sept. 2023. DOI: 10.48550/arXiv.2309.16609. arXiv: 2309.16609 [cs]. (Visited on 11/26/2025).

[13] Yuntao Bai et al. *Constitutional AI: Harmlessness from AI Feedback*. Dec. 2022. DOI: 10.48550/arXiv.2212.08073. arXiv: 2212.08073 [cs]. (Visited on 11/17/2025).

[14] Victor Barres et al. *\$τˆ2\$-Bench: Evaluating Conversational Agents in a Dual-Control Environment*. June 2025. DOI: 10.48550/arXiv.2506.07982. arXiv: 2506.07982 [cs]. (Visited on 01/26/2026).

[15] Emily M. Bender et al. "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?" In: *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*. FAccT '21. New York, NY, USA: Association for Computing Machinery, Mar. 2021, pp. 610–623. ISBN: 978-1-4503-8309-7. DOI: 10.1145/3442188.3445922. (Visited on 12/06/2025).

[16] Yoshua Bengio et al. "A Neural Probabilistic Language Model". In: *J. Mach. Learn. Res.* 3.null (Mar. 2003), pp. 1137–1155. ISSN: 1532-4435.

[17] Yoshua Bengio et al. "Managing Extreme AI Risks amid Rapid Progress". In: *Science* 384.6698 (May 2024), pp. 842–845. DOI: 10.1126/science.adn0117. (Visited on 12/07/2025).

[18] Yoshua Bengio et al. *Superintelligent Agents Pose Catastrophic Risks: Can Scientist AI Offer a Safer Path?* Feb. 2025. DOI: 10.48550/arXiv.2502.15657. arXiv: 2502.15657 [cs]. (Visited on 12/24/2025).

[19] Lukas Berglund et al. *Taken out of Context: On Measuring Situational Awareness in LLMs*. Sept. 2023. DOI: 10.48550/arXiv.2309.00667. arXiv: 2309.00667 [cs]. (Visited on 01/26/2026).

[20] Isaiah Berlin. *The Crooked Timber Of Humanity*. Random House, June 2012. ISBN: 978-1-4464-9696-1.

[21] Jan Betley et al. "Emergent Misalignment: Narrow Finetuning Can Produce Broadly Misaligned LLMs". In: *Nature* 649.8097 (Jan. 2026), pp. 584–589. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-025-09937-5. arXiv: 2502.17424 [cs]. (Visited on 01/20/2026).

[22] Nick Bostrom. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, 2014. ISBN: 978-0-19-967811-2.

[23] Nick Bostrom. "The Vulnerable World Hypothesis". In: *Global Policy* 10.4 (2019), pp. 455–476. ISSN: 1758-5899. DOI: 10.1111/1758-5899.12718. (Visited on 12/24/2025).

[24] Samuel R. Bowman et al. *Measuring Progress on Scalable Oversight for Large Language Models*. Nov. 2022. DOI: 10.48550/arXiv.2211.03540. arXiv: 2211.03540 [cs]. (Visited on 12/23/2025).

[25] RALPH ALLAN BRADLEY and MILTON E. TERRY. "RANK ANALYSIS OF INCOMPLETE BLOCK DESIGNS: THE METHOD OF PAIRED COMPARISONS". In: *Biometrika* 39.3-4 (Dec. 1952), pp. 324–345. ISSN: 0006-3444. DOI: 10.1093/biomet/39.3-4.324. (Visited on 12/15/2025).

[26] Michael Bratman. *Intention, Plans, and Practical Reason*. Cambridge, Mass. : Harvard University Press, 1987. ISBN: 978-0-674-45818-5. (Visited on 02/03/2026).

[27] Tom B. Brown et al. *Language Models Are Few-Shot Learners*. July 2020. DOI: 10.48550/arXiv.2005.14165. arXiv: 2005.14165 [cs]. (Visited on 11/26/2025).

[28] EGON BRUNSWIK. *Perception and the Representative Design of Psychological Experiments*. DGO - Digital original, 1. University of California Press, 1956. ISBN: null. DOI: 10.2307/jj.8501445. JSTOR: jj.8501445. (Visited on 02/04/2026).

[29] Sébastien Bubeck and Mark Sellke. *A Universal Law of Robustness via Isoperimetry*. Dec. 2022. DOI: 10.48550/arXiv.2105.12806. arXiv: 2105.12806 [cs]. (Visited on 12/23/2025).

[30] Patrick Butlin et al. *Consciousness in Artificial Intelligence: Insights from the Science of Consciousness*. Aug. 2023. DOI: 10.48550/arXiv.2308.08708. arXiv: 2308.08708 [cs]. (Visited on 12/07/2025).

[31] Maarten Buyl et al. *Large Language Models Reflect the Ideology of Their Creators*. Jan. 2025. DOI: 10.48550/arXiv.2410.18417. arXiv: 2410.18417 [cs]. (Visited on 01/25/2026).

[32] Donn Erwin Byrne. *The Attraction Paradigm*. New York, Academic Press, 1971. ISBN: 978-0-12-148650-1. (Visited on 01/13/2026).

[33] Stephen Casper et al. *Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback*. Sept. 2023. DOI: 10.48550/arXiv.2307.15217. arXiv: 2307.15217 [cs]. (Visited on 12/01/2025).

[34] Alan Chan et al. "Harms from Increasingly Agentic Algorithmic Systems". In: *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. FAccT '23. New York, NY, USA: Association for Computing Machinery, June 2023, pp. 651–666. ISBN: 979-8-4007-0192-4. DOI: 10.1145/3593013.3594033. (Visited on 02/03/2026).

[35] Jerry Chee et al. *QuIP: 2-Bit Quantization of Large Language Models With Guarantees*. Jan. 2024. DOI: 10.48550/arXiv.2307.13304. arXiv: 2307.13304 [cs]. (Visited on 01/25/2026).

[36] Mark Chen et al. *Evaluating Large Language Models Trained on Code*. July 2021. DOI: 10.48550/arXiv.2107.03374. arXiv: 2107.03374 [cs]. (Visited on 02/01/2026).

[37] Yanda Chen et al. *Reasoning Models Don't Always Say What They Think*. May 2025. DOI: 10.48550/arXiv.2505.05410. arXiv: 2505.05410 [cs]. (Visited on 10/28/2025).

[38] Paul Christiano. *Clarifying "AI Alignment"*. Apr. 2018. (Visited on 01/27/2026).

[39] Paul Christiano et al. *Deep Reinforcement Learning from Human Preferences*. June 2017. DOI: 10.48550/arXiv.1706.03741. arXiv: 1706.03741 [stat]. (Visited on 10/28/2025).

[40] Churchill, Winston S. *Speech to the House of Commons*. House of Commons, Nov. 1947.

[41] Ross Clark. "Selling The Family Tin? Rail Privatisation in New Zealand, in the Light of Wider Railway and Network Industry Experience". In: (Aug. 2011). (Visited on 01/19/2026).

[42] *Common Crawl*. 2025.

[43] compdemocracy. *Polis*. The Computational Democracy Project. Nov. 2025. (Visited on 11/16/2025).

[44] Ben Cottier et al. *The Rising Costs of Training Frontier AI Models*. May 2024. DOI: 10.48550/arXiv.2405.21015. arXiv: 2405.21015 [cs]. (Visited on 01/25/2026).

[45] Kate Crawford. *The Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. Yale University Press, Apr. 2021. ISBN: 978-0-300-25239-2.

[46] D. P. Crowne and D. Marlowe. "A New Scale of Social Desirability Independent of Psychopathology". In: *Journal of Consulting Psychology* 24 (Aug. 1960), pp. 349–354. ISSN: 0095-8891. DOI: 10.1037/h0047358.

[47] Robert Dale. "Sovereign AI in 2025". In: *Natural Language Processing* 31.5 (Sept. 2025), pp. 1312–1321. ISSN: 2977-0424. DOI: 10.1017/nlp.2025.10007. (Visited on 01/06/2026).

[48] David "davidad" Dalrymple et al. *Towards Guaranteed Safe AI: A Framework for Ensuring Robust and Reliable AI Systems*. July 2024. DOI: 10.48550/arXiv.2405.06624. arXiv: 2405.06624 [cs]. (Visited on 12/22/2025).

[49] Daniel Kokotajlo et al. *AI 2027*. (Visited on 07/16/2025).

[50] Daniel C. Dennett. *The Intentional Stance*. Cambridge, MA, USA: MIT Press, Mar. 1989. ISBN: 978-0-262-54053-7.

[51] Tim Dettmers et al. *LLM.Int8(): 8-Bit Matrix Multiplication for Transformers at Scale*. Nov. 2022. DOI: 10.48550/arXiv.2208.07339. arXiv: 2208.07339 [cs]. (Visited on 01/06/2026).

[52] Tim Dettmers et al. *QLoRA: Efficient Finetuning of Quantized LLMs*. May 2023. DOI: 10.48550/arXiv.2305.14314. arXiv: 2305.14314 [cs]. (Visited on 01/06/2026).

[53] Esin Durmus et al. *Towards Measuring the Representation of Subjective Global Opinions in Language Models*. Apr. 2024. DOI: 10.48550/arXiv.2306.16388. arXiv: 2306.16388 [cs]. (Visited on 12/07/2025).

[54] Joe Edelman et al. *Full-Stack Alignment: Co-Aligning AI and Institutions with Thick Models of Value*. Dec. 2025. DOI: 10.48550/arXiv.2512.03399. arXiv: 2512.03399 [cs]. (Visited on 12/17/2025).

[55] Joshua Engels et al. *Scaling Laws For Scalable Oversight*. Oct. 2025. DOI: 10.48550/arXiv.2504.18530. arXiv: 2504.18530 [cs]. (Visited on 12/23/2025).

[56] Kawin Ethayarajh et al. *KTO: Model Alignment as Prospect Theoretic Optimization*. Nov. 2024. DOI: 10.48550/arXiv.2402.01306. arXiv: 2402.01306 [cs]. (Visited on 11/26/2025).

[57] Elias Frantar et al. *GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers*. Mar. 2023. DOI: 10.48550/arXiv.2210.17323. arXiv: 2210.17323 [cs]. (Visited on 01/25/2026).

[58] Iason Gabriel. "Artificial Intelligence, Values and Alignment". In: *Minds and Machines* 30.3 (Sept. 2020), pp. 411–437. ISSN: 0924-6495, 1572-8641. DOI: 10.1007/s11023-020-09539-2. arXiv: 2001.09768 [cs]. (Visited on 12/14/2025).

[59] Aaron Gokaslan et al. *OpenWebText Corpus*. 2019.

[60] Charles Goodhart. "Problems of Monetary Management: The UK Experience". In: *Inflation, Depression, and Economic Policy in the West*. Ed. by Anthony S. Courakis. Totowa, New Jersey: Barnes and Noble Books, 1981, p. 116. ISBN: 0-389-20144-8.

[61] Ryan Greenblatt et al. *Alignment Faking in Large Language Models*. Dec. 2024. DOI: 10.48550/arXiv.2412.14093. arXiv: 2412.14093 [cs]. (Visited on 11/25/2025).

[62] Jürgen Habermas. *Between Facts and Norms: Contributions to a Discourse Theory of Law and Democracy*. Ed. by Thomas McCarthy. Trans. by William Rehg. Studies in Contemporary German Social Thought. Cambridge, MA, USA: MIT Press, Jan. 1998. ISBN: 978-0-262-58162-2.

[63] Dylan Hadfield-Menell et al. *The Off-Switch Game*. June 2017. DOI: 10.48550/arXiv.1611.08219. arXiv: 1611.08219 [cs]. (Visited on 01/19/2026).

[64] Haerpfer, Christian and Inglehart, Ronald and Moreno, Alejandro and Welzel, Christian and Kizilova, Kseniya and Diez-Medrano, Juan and Lagos, Marta and Norris, Pippa and Ponarin, Eduard and Puranen, Bi. *World Values Survey: Round Seven*. Madrid, Spain and Vienna, Austria, 2024. DOI: doi:10.14281/18241.24.

[65] Garrett James Hardin. *Filters against Folly : How to Survive despite Economists, Ecologists, and the Merely Eloquent*. New York, N.Y. : Viking, 1985. ISBN: 978-0-670-80410-8. (Visited on 02/02/2026).

[66] Dan Hendrycks, Mantas Mazeika, and Thomas Woodside. *An Overview of Catastrophic AI Risks*. Oct. 2023. DOI: 10.48550/arXiv.2306.12001. arXiv: 2306.12001 [cs]. (Visited on 12/13/2025).

[67] Dan Hendrycks et al. *Unsolved Problems in ML Safety*. June 2022. DOI: 10.48550/arXiv.2109.13916. arXiv: 2109.13916 [cs]. (Visited on 12/13/2025).

[68] Nikolaus H. R. Howe et al. "Scaling Trends in Language Model Robustness". In: *Forty-Second International Conference on Machine Learning*. 2025.

[69] Edward J. Hu et al. *LoRA: Low-Rank Adaptation of Large Language Models*. Oct. 2021. DOI: 10.48550/arXiv.2106.09685. arXiv: 2106.09685 [cs]. (Visited on 01/06/2026).

[70] Krystal Hu and Krystal Hu. "ChatGPT Sets Record for Fastest-Growing User Base - Analyst Note". In: *Reuters* (Feb. 2023). (Visited on 01/04/2026).

[71] Saffron Huang et al. "Collective Constitutional AI: Aligning a Language Model with Public Input". In: *The 2024 ACM Conference on Fairness, Accountability, and Transparency*. June 2024, pp. 1395–1417. DOI: 10.1145/3630106.3658979. arXiv: 2406.07814 [cs]. (Visited on 12/15/2025).

[72] Evan Hubinger et al. *Sleeper Agents: Training Deceptive LLMs That Persist Through Safety Training*. Jan. 2024. DOI: 10.48550/arXiv.2401.05566. arXiv: 2401.05566 [cs]. (Visited on 10/25/2025).

[73] Denis Janiak et al. *Rethinking the Evaluation of Alignment Methods: Insights into Diversity, Generalisation, and Safety*. Sept. 2025. DOI: 10.48550/arXiv.2509.12936. arXiv: 2509.12936 [cs]. (Visited on 01/25/2026).

[74] Jiaming Ji et al. *AI Alignment: A Comprehensive Survey*. Apr. 2025. DOI: 10.48550/arXiv.2310.19852. arXiv: 2310.19852 [cs]. (Visited on 12/13/2025).

[75] Jonathan Stray. *A Practical Definition of Political Neutrality for AI*. (Visited on 12/14/2025).

[76] Julia Angwin et al. *Machine Bias*. May 2016. (Visited on 01/05/2026).

[77] Daniel Kahneman and Amos Tversky. "Prospect Theory: An Analysis of Decision under Risk". In: *Econometrica : journal of the Econometric Society* 47.2 (1979), pp. 263–291.

[78] Jared Kaplan et al. *Scaling Laws for Neural Language Models*. Jan. 2020. DOI: 10.48550/arXiv.2001.08361. arXiv: 2001.08361 [cs]. (Visited on 12/24/2025).

[79] Zachary Kenton et al. *On Scalable Oversight with Weak LLMs Judging Strong LLMs*. July 2024. DOI: 10.48550/arXiv.2407.04622. arXiv: 2407.04622 [cs]. (Visited on 12/23/2025).

[80] Hannah Rose Kirk et al. *The Empty Signifier Problem: Towards Clearer Paradigms for Operationalising "Alignment" in Large Language Models*. Nov. 2023. DOI: 10.48550/arXiv.2310.02457. arXiv: 2310.02457 [cs]. (Visited on 01/27/2026).

[81] Oliver Klingefjord, Ryan Lowe, and Joe Edelman. *What Are Human Values, and How Do We Align AI to Them?* Apr. 2024. DOI: 10.48550/arXiv.2404.10636. arXiv: 2404.10636 [cs]. (Visited on 12/16/2025).

[82] Anja Kollmuss and Julian Agyeman. "Mind the Gap: Why Do People Act Environmentally and What Are the Barriers to pro-Environmental Behavior?" In: *Environmental Education Research* 8.3 (Aug. 2002), pp. 239–260. ISSN: 1350-4622. DOI: 10.1080/13504620220145401. (Visited on 12/14/2025).

[83] S. Kullback and R. A. Leibler. "On Information and Sufficiency". In: *The Annals of Mathematical Statistics* 22.1 (Mar. 1951), pp. 79–86. ISSN: 0003-4851, 2168-8990. DOI: 10.1214/aoms/1177729694. (Visited on 09/16/2025).

[84] Thomas Kwa et al. *Measuring AI Ability to Complete Long Tasks*. Mar. 2025. DOI: 10.48550/arXiv.2503.14499. arXiv: 2503.14499 [cs]. (Visited on 12/24/2025).

[85] Harrison Lee et al. *RLAIF vs. RLHF: Scaling Reinforcement Learning from Human Feedback with AI Feedback*. Sept. 2024. DOI: 10.48550/arXiv.2309.00267. arXiv: 2309.00267 [cs]. (Visited on 11/26/2025).

[86] Pola Lehmann et al. *The Manifesto Data Collection. Manifesto Project (MRG/CMP/MARPOR). Version 2025a*. 2025. DOI: 10.25522/manifesto.mpds.2025a.

[87] Jan Leike et al. *Scalable Agent Alignment via Reward Modeling: A Research Direction*. Nov. 2018. DOI: 10.48550/arXiv.1811.07871. arXiv: 1811.07871 [cs]. (Visited on 12/23/2025).

[88] Leon Festinger. *A Theory of Cognitive Dissonance*. 1957. (Visited on 12/15/2025).

[89] Cheng Li et al. *CultureLLM: Incorporating Cultural Differences into Large Language Models*. Dec. 2024. DOI: 10.48550/arXiv.2402.10946. arXiv: 2402.10946 [cs]. (Visited on 12/07/2025).

[90] Xuechen Li et al. *AlpacaEval: An Automatic Evaluator of Instruction-Following Models*. May 2023.

[91] Yunxiang Li et al. *ChatDoctor: A Medical Chat Model Fine-Tuned on a Large Language Model Meta-AI (LLaMA) Using Medical Domain Knowledge*. June 2023. DOI: 10.48550/arXiv.2303.14070. arXiv: 2303.14070 [cs]. (Visited on 12/06/2025).

[92] Ji Lin et al. *AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration*. July 2024. DOI: 10.48550/arXiv.2306.00978. arXiv: 2306.00978 [cs]. (Visited on 01/25/2026).

[93] Xiao Liu et al. *AgentBench: Evaluating LLMs as Agents*. Oct. 2025. DOI: 10.48550/arXiv.2308.03688. arXiv: 2308.03688 [cs]. (Visited on 01/26/2026).

[94] Jiasen Lu et al. *ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations for Vision-and-Language Tasks*. Aug. 2019. DOI: 10.48550/arXiv.1908.02265. arXiv: 1908.02265 [cs]. (Visited on 12/07/2025).

[95] Aditya Malte and Pratik Ratadiya. *Evolution of Transfer Learning in Natural Language Processing*. Oct. 2019. DOI: 10.48550/arXiv.1910.07370. arXiv: 1910.07370 [cs]. (Visited on 01/05/2026).

[96] David Manheim. "The Fragile World Hypothesis: Complexity, Fragility, and Systemic Existential Risk". In: *Futures* 122 (Sept. 2020), p. 102570. ISSN: 0016-3287. DOI: 10.1016/j.futures.2020.102570. (Visited on 12/24/2025).

[97] Alexander Meinke et al. *Frontier Models Are Capable of In-context Scheming*. Jan. 2025. DOI: 10.48550/arXiv.2412.04984. arXiv: 2412.04984 [cs]. (Visited on 11/16/2025).

[98] Stephen Merity, Nitish Shirish Keskar, and Richard Socher. *Regularizing and Optimizing LSTM Language Models*. Aug. 2017. DOI: 10.48550/arXiv.1708.02182. arXiv: 1708.02182 [cs]. (Visited on 12/06/2025).

[99] METR. *Measuring AI Ability to Complete Long Tasks*. Mar. 2025.

[100] Silvia Milano, Mariarosaria Taddeo, and Luciano Floridi. "Recommender Systems and Their Ethical Challenges". In: *AI & SOCIETY* 35.4 (Dec. 2020), pp. 957–967. ISSN: 1435-5655. DOI: 10.1007/s00146-020-00950-y. (Visited on 12/22/2025).

[101] John Stuart Mill. *On Liberty*. Jan. 2011. (Visited on 12/22/2025).

[102] Prakash Nadkarni. "Chapter 4 - Core Technologies: Machine Learning and Natural Language Processing". In: *Clinical Research Computing*. Ed. by Prakash Nadkarni. Academic Press, 2016, pp. 85–114. ISBN: 978-0-12-803130-8. DOI: 10.1016/B978-0-12-803130-8.00004-X.

[103] john Von Neumann. *Theory Of Games And Economic Behavior*. 1944. (Visited on 01/26/2026).

[104] nfelt. *Sovereign Remedies: Between AI Autonomy and Control*. Apr. 2025. (Visited on 01/06/2026).

[105] Shangrui Nie et al. *Survey-to-Behavior: Downstream Alignment of Human Values in LLMs via Survey Questions*. Aug. 2025. DOI: 10.48550/arXiv.2508.11414. arXiv: 2508.11414 [cs]. (Visited on 12/02/2025).

[106] Richard E. Nisbett and Timothy D. Wilson. "Telling More Than We Can Know: Verbal Reports on Mental Processes". In: *Psychological Review* 84.3 (1977), pp. 231–59. DOI: 10.1037/0033-295x.84.3.231.

[107] Ziad Obermeyer et al. "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations". In: *Science (New York, N.Y.)* 366.6464 (Oct. 2019), pp. 447–453. ISSN: 1095-9203. DOI: 10.1126/science.aax2342.

[108] Stephen M. Omohundro. "The Basic AI Drives". In: *Artificial Intelligence Safety and Security*. Ed. by Roman V. Yampolskiy. First edition. — Boca Raton, FL : CRC Press/Taylor & Francis Group, 2018.: Chapman and Hall/CRC, July 2018, pp. 47–55. ISBN: 978-1-351-25138-9. DOI: 10.1201/9781351251389-3. (Visited on 01/19/2026).

[109] Long Ouyang et al. *Training Language Models to Follow Instructions with Human Feedback*. Mar. 2022. DOI: 10.48550/arXiv.2203.02155. arXiv: 2203.02155 [cs]. (Visited on 10/25/2025).

[110] Aviv Ovadya et al. *Democratic AI Is Possible. The Democracy Levels Framework Shows How It Might Work*. Aug. 2025. DOI: 10.48550/arXiv.2411.09222. arXiv: 2411.09222 [cs]. (Visited on 01/04/2026).

[111] Alexander Pan et al. "Do the Rewards Justify the Means? Measuring Trade-Offs between Rewards and Ethical Behavior in the Machiavelli Benchmark." In: *ICML* (2023).

[112] *Pause Giant AI Experiments: An Open Letter*. (Visited on 12/24/2025).

[113] *PauseAI Proposal*. https://pauseai.info/proposal. (Visited on 12/24/2025).

[114] Long Phan et al. *Humanity's Last Exam*. 2025. arXiv: 2501.14249 [cs.LG].

[115] Philip Gage. "A New Algorithm for Data Compression". In: *The C Users Journal* FEB94 (Feb. 1994). (Visited on 12/06/2025).

[116] Steven Pinker. *The Better Angels of Our Nature*. Oct. 2012. (Visited on 01/27/2026).

[117] Philip M. Podsakoff et al. "Common Method Biases in Behavioral Research: A Critical Review of the Literature and Recommended Remedies". In: *The Journal of Applied Psychology* 88.5 (Oct. 2003), pp. 879–903. ISSN: 0021-9010. DOI: 10.1037/0021-9010.88.5.879.

[118] Public AI. *Public AI White Paper*. Aug. 2024. DOI: 10.5281/zenodo.13914560. (Visited on 11/17/2025).

[119] Xiangyu Qi et al. *Safety Alignment Should Be Made More Than Just a Few Tokens Deep*. June 2024. DOI: 10.48550/arXiv.2406.05946. arXiv: 2406.05946 [cs]. (Visited on 12/01/2025).

[120] R. H. Coarse. "The Problem of Social Cost: The Journal of Law and Economics: Vol 3". In: *The Journal of Law and Economics* (). (Visited on 02/02/2026).

[121] Alec Radford et al. "Language Models Are Unsupervised Multitask Learners". In: 2019. (Visited on 01/25/2026).

[122] Rafael Rafailov et al. *Direct Preference Optimization: Your Language Model Is Secretly a Reward Model*. July 2024. DOI: 10.48550/arXiv.2305.18290. arXiv: 2305.18290 [cs]. (Visited on 10/25/2025).

[123] *Report of the Detailed Findings of the Independent International Fact-finding Mission on Myanmar*. UN Fact-Finding Mission Report A/HRC/39/CRP.2. United Nations Human Rights Council, Independent International Fact-Finding Mission on Myanmar, 2018-09-17, 2018.

[124] F. J. Roethlisberger. *Management and the Worker*. Cambridge, Mass. : Harvard Uuniversity Press, 1939. (Visited on 02/05/2026).

[125] Milton Rokeach. *The Nature of Human Values*. Free Press, 1973. ISBN: 978-0-02-926750-9.

[126] David Ross. *The Right and the Good*. Ed. by Philip Stratton-Lake. Oxford University Press, Aug. 2002. ISBN: 978-0-19-925265-7. DOI: 10.1093/0199252653.001.0001. (Visited on 01/27/2026).

[127] Stuart Russell and Peter Norvig. *Artificial Intelligence, Global Edition*. Pearson Education, 2021. ISBN: 978-1-292-40117-1. (Visited on 01/20/2026).

[128] Paul A. Samuelson. "The Pure Theory of Public Expenditure". In: *The Review of Economics and Statistics* 36.4 (1954), pp. 387–389. ISSN: 0034-6535. DOI: 10.2307/1925895. JSTOR: 1925895. (Visited on 02/02/2026).

[129] A. Sandberg and N. Bostrom. "Whole Brain Emulation: A Roadmap". In: (2008). (Visited on 12/23/2025).

[130] Jonas B. Sandbrink. *Artificial Intelligence and Biological Misuse: Differentiating Risks of Language Models and Biological Design Tools*. Dec. 2023. DOI: 10.48550/arXiv.2306.13952. arXiv: 2306.13952 [cs]. (Visited on 01/05/2026).

[131] Bruce Schneier Sanders Nathan E. *Build AI by the People, for the People*. June 2023. (Visited on 12/23/2025).

[132] Shibani Santurkar et al. *Whose Opinions Do Language Models Reflect?* Mar. 2023. DOI: 10.48550/arXiv.2303.17548. arXiv: 2303.17548 [cs]. (Visited on 12/03/2025).

[133] Leonard J. Savage. *The Foundations of Statistics*. New York, Wiley, 1954. (Visited on 01/26/2026).

[134] Timo Schick et al. *Toolformer: Language Models Can Teach Themselves to Use Tools*. Feb. 2023. DOI: 10.48550/arXiv.2302.04761. arXiv: 2302.04761 [cs]. (Visited on 01/20/2026).

[135] John Schulman et al. *Proximal Policy Optimization Algorithms*. Aug. 2017. DOI: 10.48550/arXiv.1707.06347. arXiv: 1707.06347 [cs]. (Visited on 03/11/2025).

[136] Schwartz. "Universals in the Content and Structure of Values: Theoretical Advances and Empirical Tests in 20 Countries". In: *Advances in Experimental Social Psychology*. Vol. 25. Academic Press, Jan. 1992, pp. 1–65. DOI: 10.1016/S0065-2601(08)60281-6. (Visited on 12/12/2025).

[137] Shalom Schwartz. "An Overview of the Schwartz Theory of Basic Values". In: *Online Readings in Psychology and Culture* 2 (Dec. 2012). DOI: 10.9707/2307-0919.1116.

[138] Shalom H. Schwartz. "Are There Universal Aspects in the Structure and Contents of Human Values?" In: *Journal of Social Issues* 50.4 (1994), pp. 19–45. ISSN: 1540-4560. DOI: 10.1111/j.1540-4560.1994.tb01196.x. (Visited on 12/05/2025).

[139] Shalom H. Schwartz et al. “Refining the Theory of Basic Individual Values”. In: *Journal of Personality and Social Psychology* 103.4 (Oct. 2012), pp. 663–688. ISSN: 1939-1315. DOI: 10.1037/a0029393.

[140] Zhihong Shao et al. *DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models*. Apr. 2024. DOI: 10.48550/arXiv.2402.03300. arXiv: 2402.03300 [cs]. (Visited on 11/26/2025).

[141] Shoshana Zuboff. *The Age of Surveillance Capitalism*. PublicAffairs, June 2017. ISBN: 978-1-61039-569-4. (Visited on 01/05/2026).

[142] Herbert A. Simon. “A Behavioral Model of Rational Choice”. In: *The Quarterly Journal of Economics* 69.1 (1955), pp. 99–118. ISSN: 0033-5533. DOI: 10.2307/1884852. JSTOR: 1884852. (Visited on 01/26/2026).

[143] Peter Singer. *The Expanding Circle : Ethics and Sociobiology*. Oxford : Clarendon Press, 1981. ISBN: 978-0-19-824646-6. (Visited on 01/27/2026).

[144] Nate Soares et al. “Corrigibility”. In: *AI and Ethics*. 2015. (Visited on 01/19/2026).

[145] “Speculations Concerning the First Ultraintelligent Machine”. In: *Advances in Computers*. Vol. 6. Elsevier, Jan. 1966, pp. 31–88. DOI: 10.1016/S0065-2458(08)60418-0. (Visited on 12/24/2025).

[146] Joseph E. Stiglitz. “Markets, Market Failures, and Development”. In: 79.2 (1989), pp. 197–203. DOI: 10.7916/D8BK1PD1. (Visited on 12/22/2025).

[147] Matt Stroud. *An Automated Policing Program Got This Man Shot Twice*. https://www.theverge.com/c/2244 listed-csk-entry. May 2021. (Visited on 01/05/2026).

[148] Joseph Suh et al. “Language Model Fine-Tuning on Scaled Survey Data for Predicting Distributions of Public Opinions”. In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Wanxiang Che et al. Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 21147–21170. ISBN: 979-8-89176-251-0. DOI: 10.18653/v1/2025.acl-long.1028. (Visited on 12/03/2025).

[149] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning, Second Edition: An Introduction*. MIT Press, Nov. 2018. ISBN: 978-0-262-35270-3.

[150] “Tackling the Perils of Dual Use in AI”. In: *Nature Machine Intelligence* 4.4 (Apr. 2022), pp. 313–313. ISSN: 2522-5839. DOI: 10.1038/s42256-022-00484-6. (Visited on 01/05/2026).

[151] Gemini Team et al. “Gemini: A Family of Highly Capable Multimodal Models”. In: *arXiv preprint arXiv:2312.11805* (2023). arXiv: 2312.11805.

[152] GLM-4 5 Team et al. *GLM-4.5: Agentic, Reasoning, and Coding (ARC) Foundation Models*. Aug. 2025. DOI: 10.48550/arXiv.2508.06471. arXiv: 2508.06471 [cs]. (Visited on 01/25/2026).

[153] *The Open Source AI Definition - 1.0 - Open Source Initiative*. https://opensource.org/ai/open-source-ai-definition. (Visited on 01/25/2026).

[154] Cameron Tice et al. *Alignment Pretraining: AI Discourse Causes Self-Fulfilling (Mis)Alignment*. Jan. 2026. DOI: 10.48550/arXiv.2601.10160. arXiv: 2601.10160 [cs]. (Visited on 01/21/2026).

[155] Alexis de Tocqueville and Henry Reeve. *Democracy in America*. London: Saunders and Otley, 1835. (Visited on 12/22/2025).

[156] Tomas Mikolav. "STATISTICAL LANGUAGE MODELS BASED ON NEURAL NETWORKS". PhD thesis. BRNO UNIVERSITY OF TECHNOLOGY. (Visited on 12/06/2025).

[157] Amos Tversky and Daniel Kahneman. "Advances in Prospect Theory: Cumulative Representation of Uncertainty". In: *Journal of Risk and Uncertainty* 5.4 (Oct. 1992), pp. 297–323. ISSN: 1573-0476. DOI: 10.1007/BF00122574. (Visited on 12/01/2025).

[158] Amos Tversky and Daniel Kahneman. "Judgment under Uncertainty: Heuristics and Biases". In: *Science* 185.4157 (Sept. 1974), pp. 1124–1131. DOI: 10.1126/science.185.4157.1124. (Visited on 01/26/2026).

[159] University of Auckland. *New Zealand Attitudes and Values Study (NZAVS), 2009–2025*. 2025.

[160] Tyler J. VanderWeele. "On the Promotion of Human Flourishing". In: *Proceedings of the National Academy of Sciences* 114.31 (Aug. 2017), pp. 8148–8156. DOI: 10.1073/pnas.1702996114. (Visited on 12/22/2025).

[161] Hal R. Varian. *Intermediate Microeconomics: A Modern Approach*. W.W. Norton & Company, 2010. ISBN: 978-0-393-93424-3.

[162] Vast.ai, Inc. *Vast.Ai – Affordable GPU Cloud Marketplace*. https://cloud.vast.ai/. 2026. (Visited on 01/07/2026).

[163] Ashish Vaswani et al. *Attention Is All You Need*. 2017. DOI: 10.48550/arXiv.1706.03762. arXiv: 1706.03762 [cs]. (Visited on 12/06/2025).

[164] Guanzhi Wang et al. *Voyager: An Open-Ended Embodied Agent with Large Language Models*. Oct. 2023. DOI: 10.48550/arXiv.2305.16291. arXiv: 2305.16291 [cs]. (Visited on 01/20/2026).

[165] Jason Wei et al. *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*. Jan. 2023. DOI: 10.48550/arXiv.2201.11903. arXiv: 2201.11903 [cs]. (Visited on 12/06/2025).

[166] Mark Wenman. *Agonistic Democracy: Constituent Power in the Era of Globalisation*. Cambridge: Cambridge University Press, 2013.

[167] Norbert Wiener. *The Human Use of Human Beings: Cybernetics and Society*. HarperCollins, Aug. 2025. ISBN: 978-0-06-342320-6.

[168] Guangxuan Xiao et al. *SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models*. Mar. 2024. DOI: 10.48550/arXiv.2211.10438. arXiv: 2211.10438 [cs]. (Visited on 01/25/2026).

[169] Shunyu Yao et al. *ReAct: Synergizing Reasoning and Acting in Language Models*. Mar. 2023. DOI: 10.48550/arXiv.2210.03629. arXiv: 2210.03629 [cs]. (Visited on 01/20/2026).

[170] Wenlong Zhao et al. "WorldValuesBench: A Large-Scale Benchmark Dataset for Multi-Cultural Value Awareness of Language Models". In: *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. Ed. by Nicoletta Calzolari et al. Torino, Italia: ELRA and ICCL, May 2024, pp. 17696–17706. (Visited on 12/07/2025).

[171] Lianmin Zheng et al. *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. Dec. 2023. DOI: 10.48550/arXiv.2306.05685. arXiv: 2306.05685 [cs]. (Visited on 01/26/2026).

[172] Chunting Zhou et al. *LIMA: Less Is More for Alignment*. May 2023. DOI: 10.48550/arXiv.2305.11206. arXiv: 2305.11206 [cs]. (Visited on 02/02/2026).

Appendix A

Related ideas

A.1 Large Language Models

Here I provide a brief overview of large language models (LLMs), their architecture, training methods, and capabilities. This is not meant to be an exhaustive review, but rather a primer to understand the context of LLM alignment.

A.1.1 What are Language Models?

Language models (LMs) are a class of machine learning models that are designed to understand and generate human language. They are typically probabilistic models that learn to predict the next token in a sequence of text given the preceding context. By training on large datasets of text data (corpora), language models can learn the statistical patterns and structures inherent in human language, enabling them to generate coherent and contextually relevant text.

Language models see the world as **tokens** which are generated by taking a string and chunking it into smaller pieces. Tokens can represent words, parts of words, or even individual characters depending on the **tokenization** method used. Most language models today use **Byte pair encoding** [115] that breaks text into subwords. Almost all successful language models use probabilistic methods, therefore they are models of the form $P(t_i|t_1, t_2, ..., t_{i-1})$ where t_i is the token to be predicted and $t_1, t_2, ..., t_{i-1}$ are the preceding tokens in the sequence. Earlier methods of rule based systems failed due to the complexity and ambiguity of human language and rules being too rigid [102, 95].

A.1.2 Early probabilistic language models

Early probabilistic models included simple statistical approaches such as n-grams, which relied on counting the frequency of token sequences. However this approach suffered from two competing issues, short context windows (2-grams, 3-grams etc) means that model can't understand long range dependencies in text, while larger n-grams leads to data sparsity issues where many long sequences are never seen in training data. The advent of neural networks allowed for more sophisticated models such as **Feedforward Neural Language models** [16], **Recurrent Neural Networks (RNNs)** [156] and **Long Short-Term Memory networks (LSTMs)** [98] which could in theory capture longer range dependencies in text. However these models suffered from issues such as

- **Vanishing gradients**: Early tokens in a long sequence have little effect on gradients and therefore the models understanding of long range dependencies.

- **Information bottlenecks**: The hidden state of the model has to compress all prior context into a fixed size vector.

Crucially these models failed in their flexibility to model the complex dependencies in human language.

A.1.3 Modern attention based Large Language Models

The key feature in language is context, and it is this context of surrounding words that let us humans understand and break through the complexity and ambiguity of language. The challenge is that the way in which a words company effects the meaning of the word is complex, therefore any language model needs to be able effectively capture these complex relationships. **Attention** is the concept where each token's representation is influenced by all the other tokens in the sequence (or just the preceding tokens in the case of **autoregressive** models). This was first applied to RNN-based LM in [11], however in [163] it was shown that attention mechanisms alone were sufficient to model language without the need for any recurrent structure[1].

Transformers [163] are a class of neural network **architecture** that utilizes the attention mechanism to process data and generate a concise representation (**encoder**-only), or generate data autoregressively (**decoder**-only). The canonical transformer architecture consists of two parts an encoder and a decoder, however decoder only architectures are shown to be sufficient for all current generative language modelling tasks. A simplified architecture of a decoder-only transformer model is shown in Figure A.1. The key feature of the transformer is the **self attention**. Each token has an **embedding vector** (used to represent its context aware meaning, i.e river *bank* vs *bank* robbery) which is iteratively updated by each layer of self-attention. The self-attention mechanism involves matrix multiplication between the current token and other tokens in the sequence. For generative tasks an attention mask is applied so that each token can only 'see' preceding tokens in the sequence. Furthermore to allow for different types of dependencies to be captured this attention is done multi times in parallel (**multi-head attention**) and merged together at the end of each layer. After several layers of self-attention the final set of token embeddings are used to generate a probability distribution over the vocabulary for the next token. A token can be sampled from this distribution, appended to the input sequence and the process repeated to generate long sequences of text.

Modern transformer based large language models (LLMs) such as GPT-3 [27] are some of the largest machine learning models ever created with hundreds of billions of parameters and now trillions of parameters. Therefore training these models requires massive datasets. They are trained in a self supervised manner on large corpora (hundreds of terabytes) of text data from the Internet such as Common Crawl [42], WebText [59], and others. The objective function is usually the cross-entropy loss between the predicted token distribution and the actual next token in the sequence. The output of this training process is a model that can generate coherent and contextually relevant text given a prompt.

A.1.4 Scale of LLMs

A LLM is a specialized type of **Artificial Neural Network (ANN)**. ANNs are a class of machine learning models inspired by the structure and function of biological neural networks (i.e our brain). Where a bunch of **neurons** are connected to gather and form a network that processes

[1] Recurrent structure is where there is a hidden state (normally a fixed length vector) that is passed along and updated each time by the next input. For language modelling this hidden state is supposed to contain the context needed to understand the next input token. However as discussed above it was not sufficient.

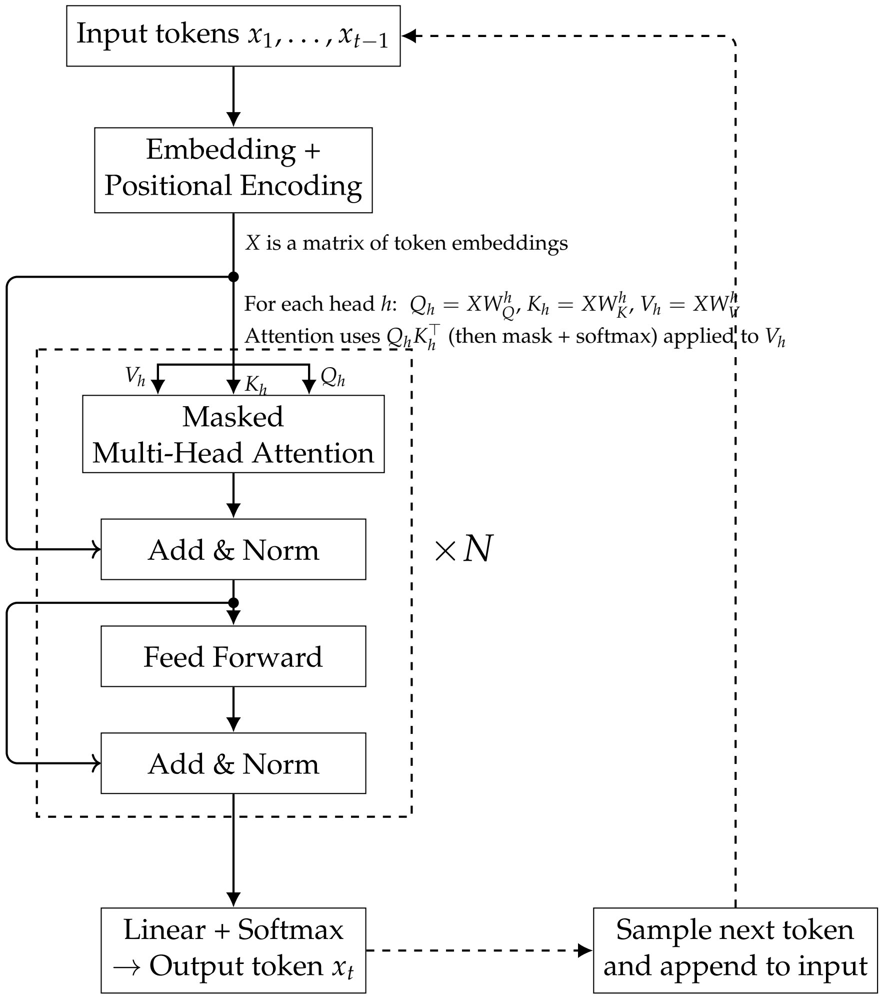

Figure A.1: Simplified architecture of a decoder-only transformer model for language modeling. Adapted from [163].

information. The key design decisions in a ANN are the architecture (which neurons are connected to which) and the **parameters** (the weights of the connections between neurons). As discussed above most modern LLMs use derivatives of the transformer architecture. They operate on a scale that is unprecedented in machine learning. When GPT-2 was released in 2019 it has 1.5 billion parameters [121] GPT-3 in 2020 had 175 billion parameters [27] and current frontier models have in the trillions of parameters.

The scale of these models is important for two reasons. Firstly is that scale alone was found to improve the capabilities of LLMs [78]. This is what drove the rapid increase in model size and training data over the last few years, where adding more compute was more effective than theoretical advances. Secondly is that the scale of these models means that training them requires massive computational resources. A frontier model with hundreds of billions of parameters is estimated to cost in the order of hundreds of millions to billions of dollars to train [44]. With the rise in model and dataset size there is an increasing situation where only a few very well funded organizations can train these models.

Fortunately there is an Open Source AI movement [153] which pushes for trained models along with the training process and data are shared publicly. This has meant that there are various "open source" LLMs that reach similar capabilities to proprietary models [140, 152] [2]. With these open weight models it is possible for smaller groups to fine tune them for specific tasks or domains at a fraction of the cost of training a new model from scratch. There are two key trends that can make use of these open models more effective. The first is **parameter efficient fine-tuning** methods [69] that allow for effective fine-tuning of large models by only updating a small fraction of the model parameters. The second is model **quantization** methods [52, 57, 168, 92, 35] that allow for the models parameters to be stored and computed in lower precisions formats (such as 4-bit instead of 16 or 32-bit) which drastically reduces the memory and compute requirements needed to run these models.

A.1.5 LLMs in the modern world

Large language models (LLMs) are fundamentally just language models which can predict the next token in a sequence of text. However due to the prevalence of natural language in our world and the capabilities of LLMs they have applications in a wide variety of domains. This includes Virtual Assistants, Content creation, Code generation, Translation and many more. To understand their capabilities one can look at current benchmarks and LLMs ability like winning math olympiads, passing the bar exam. The meteoric rise of capabilities has lead to problems with not having tests that are sufficiently hard [114] along with observations that on current trajectories they will outpace human thinking within the next few years [49, 99]. Additionally the transformer architecture has been demonstrated to be effective in other domains like computer vision [94] which has lead to **multi-modal models** that can process and generate text, images, and other data types [2]. This flexibility of LLMs have lead to them have widespread adoption and proliferation in modern information systems [3]. The abilities of the AI systems is such that we are seeing them employed within highly capable **AI agents** that perceive their environment, make plans, and act autonomously to achieve goals [169, 164, 134, 127].

Along with this proliferation there has been increasing concern about the safety and ethical implications of these models. Specifically these models have been shown to exhibit concerning behaviors like self-preservation and deception [97, 61, 37]. On top of this there

[2]The open source movement advocates for complete transparency that includes all that would be needed to recreate the model yourself. Most of the open LLMs are simply open weight models that can have restrictive licenses.

[3]ChatGPT which was the first widely used LLM was the fastest growing website ever[70].

are growing ethical questions around its use in our society [15, 17] as well as concerns of the welfare of the models themselves [30].

A.2 Challenges beyond value alignment

AI alignment is a crucial step towards safe and beneficial AI systems. There are problems that are adjacent to alignment within AI safety that need to be addressed to ensure that the introduction of AI systems into society is a net positive.

A.2.1 Concrete problems of AI safety

One can think of AI systems in an abstract way and outline the key problems that are always present when working with AI systems. There have been multiple attempts in the last decade to outline and categorize these problems [4, 67]. I will outline some of the key problems here in a pseudo-categorization that can be understood as: always safe, capable and overseen.

Firstly is **robustness** that ensures the model behaves reasonably at all times, specifically in situations that are out of distribution (adversarial, black swan, shifts etc) from the training data. A key challenge to this is that models mostly are unaware of when they are out of distribution therefore they must "always behave reasonably". This is easily implemented in a vacuum robot where if it is sufficiently confused it can just stop moving, however it is unclear how to implement this level of robustness in an AI doctor or lawyer as shutting down is not a reasonable response to a difficult situation. A promising direction to solve this problem is demonstrated phenomena that increasing model complexity increases its robustness [29]. At a high level this can be intuitive as a more complex model can contain and store more information about the world and therefore be able to generalize better to new situations. There has been limited evidence that this is true in modern LLM [68], with larger models having slightly more base robustness and being much more effective at receiving robustness training. Current models are still orders of magnitude away from the complexity of the human brain [129] and so may have much more "free" robustness to gain as they scale up in size and complexity.

Secondly is **factual accuracy and hallucinations** which is ensuring that the model has a correct model of the world and also a correct model of itself and its capabilities. This is particularly crucial with LLMs as they generate text which has the ability of being perfectly well written, plausible, yet completely false. This is because the model fundamentally isn't in tune with the universe and so can have no concept of when it is delusional. This is much the same problem that humans have, and we rely on external signals from society to guide our own model of how sane we are.

Lastly is **scalable oversight** which is ensuring we can keep track of and monitor the behavior of AI systems as they both scale up in capability and scale out in number of systems. If we had a provably aligned and robust system then oversight is not needed, however it is unlikely such a system will exist, instead we will likely have "demonstrably" aligned and robust systems. The only way to maintain confidence that these systems are behaving well is to continually monitor them. Doing this when they are highly capable presents several challenges due to how effectively they could deceive and/or how poorly humans could even understand what they are doing. Equally so when there are a large number of systems the available resources to monitor each agent decreases and so increasingly efficient and effective monitoring techniques will be needed. There are some promising results of bootstrap supervision [24, 55, 79] where AI systems are used to help monitor other AI systems. Although potentially quite fruitful this approach introduces new challenges of misalignment propagation as one needs to ensure that the supervising AI system is also robust and aligned.

A.2.2 Current negative impacts of AI

AI is already in the world and having an effect. There are many different types of AI harms in the near term. Common themes of negative impacts of AI are as follows:

- **Bias and discrimination:** AI systems can perpetuate and amplify existing biases in society. For example, in predictive policing [147], judicial sentencing [76], and hiring decisions.
- **Automation and job loss:** AI systems can automate tasks that were previously done by humans, leading to job loss and economic disruption [10]. This can disproportionately affect certain groups of people, such as low-skilled workers and marginalized communities. Furthermore this leads to increased wealth concentration as the owners of companies that deploy AI systems reap the majority of the benefits.
- **Safety harms in high risk areas:** AI systems deployed in high-risk areas such as healthcare [107], transportation, and critical infrastructure can lead to real-world safety harms if they fail or make incorrect decisions.
- **Population manipulation:** AI systems like recommender systems, online advertising and social media algorithms can manipulate public opinion and behavior [141, 3]. This has had some severe real-world consequences, such as the Rohingya genocide in Myanmar [123].
- **Explicit misuse:** More capable AI systems are at greater risk of being used for malicious purposes, which is called the dual-use problem. Common dual-use concerns include bioweapon design and cyber warfare [150, 130].

Many of these harms come about due to either negligent or malicious deployment of AI systems. Having systems that are better aligned to human values could help mitigate some of these harms. However there are also systemic issues with the way that AI is developed and deployed that lead to these harms. Namely the profit motive of private companies is not aligned with the goal of ”helping users being their best self”. Therefore solutions to these problems are the same solutions to how we stop companies in general from causing harm, namely regulation, oversight and changing the incentives of companies. This can be effectively done using the principles of public democratic AI outlined in 2.4.

A.2.3 Catastrophic and existential risks of AI

The stakes are high with AI. There are multiple plausible scenarios where AI could cause human catastrophic or existential risks to humanity. We may be living in a very vulnerable world where the majority of possible future states are negative from our current perspectives [23, 96]. Central to these paths and scenarios is the idea that more powerful and capable AI systems have the potential to enfeeble humans and disempower us. Once in this state of disempowerment, humans are by definition unable to control and direct their own future. Therefore, if any form of misalignment exists between the AI system and the maximum human flourishing goal, the consequences could be **catastrophic** or **existential**

AI development and AI progress has increased rapidly in the last few years [84]. It has increased in speed due to scaling laws of AI [78] and increased investment from both private and public sector. The key next step is that AI systems will accelerate and automate many parts of AI research. This could result in an intelligence explosion [22, 145] where the input that humans play in AI development tends towards zero and improvement can happen on timescales much faster than humans. Even if this intelligence explosion happens over the course of many years it is still much faster than human development over the last few thousand years. Therefore the probability of highly capable AI systems could be much higher than previously thought assuming AI can appreciably automate its own development.

With these highly capable AI systems there are several plausible catastrophic risks that could occur. These can generally be split up into two sorts of catastrophic risk, **rogue AI** and **malicious use of AI** [66]. Rogue AI is where a system is either deceptively, overtly or indifferently doing actions against the goal of maximizing median human flourishing. The impact of this risk increases as the capabilities and proliferation of AI systems increases. The second risk also has similar increasing impact, where bad actors can do increasingly harmful things with increasingly powerful and accessible systems (bioweapons, cyber warfare etc). Either of these situations have the potential to cause an irreversible catastrophic risk to humanity. This could be through direct extinction of humanity, irreversible collapse of civilization or permanent disempowerment of humanity [22].

To compound these risks further are the **instrumental goals** that agents are likely to pursue to achieve their final goals [18, 108]. There are three relevant instrumental goals. The first is **power seeking** in which any agent that wants to complete a task will be able to increase its probability of success by gaining more power and resources [108]. The second is **self-preservation** where being turned off will reduce the probability of success and therefore agents will try to avoid being turned off [63]. The third is **goal-content integrity** where for an agent to achieve its goals it must ensure that its goals don't change [144]. From these instrumental goals one could form a argument that an active intelligent system (even if perfectly aligned to maximum human flourishing) could still cause harm, much more so than an mis aligned yet much less capable system.

As the impact of these risks increases the probability of them can also be increased through both organizational incompetence (e.g. The USSR and Chernobyl, NASA and the Challenger disaster) and competitive pressures (e.g. nuclear arms race) [66]. These two factors broadly lead to decreased safety standards with increased focus on rapid development and deployment.

A.2.4 Preventing catastrophic and existential risks of AI

Given that the stakes are very high, there should be significant effort to ensure that these risks are mitigated and negative events avoided. To prevent the risks of AI, we can either try to slow down (restrictive) or try to make it safer (constructive).

The first common approach is a halt or moratorium on AI development. Although reasonable sounding at first glance relatively large scale attempts at this have failed [112]. A global pause struggles from coordination problem where each actor has no incentive to pause as others might not pause and therefore gain competitive advantage. Furthermore pausing in an indefinite manner runs into the ethical problem about forgoing potential benefits of AI [4]. This is why most pauses call for a limited time period to allow for safety, policy and governance research and implementations to catch up [113]. Given the nature of AI development requiring large amounts of resources (hardware, data, power, etc) it is plausible that a well designed pause and global coordination could be effective. It could work much like nuclear non-proliferation treaties where countries agree to limit their development of nuclear weapons and allow for inspections to ensure compliance. Also much like nuclear weapons most of current day frontier AI development requires massive data centres that are hard to hide and therefore easier to monitor, much like uranium refinement factories.

[4] This concept of whether we do need AI and the importance of bringing it to fruition compared to its potential downsides is a very important question that shouldn't be glossed over too much. However in the context of AI alignment it is outside the scope and thus not discussed further.

If one had time what would be the methods and ways to reduce the risks? Generally speaking they fall into technical safety solutions and governance and policy solutions. There are proposals for having various policies and regulations that would slow down the progress of AI development. Namely so that we have more time to understand the systems and develop appropriate safety measures. Policies for slowdown can involve computation restrictions, increased liability for the developers and sellers of AI systems as well as more stringent testing and evaluation requirements. Some of these can help long term safety however most of them are simply mandating a slower process and better appreciation of the risks.

With this better appreciation of the risks we can implement further policies and also develop novel technical solutions. There are various methods for AI that are safe by design [48, 18]. The key to these methods is that the resultant AI is either deeply aligned by construction or is limited in capability to prevent misalignment from causing catastrophic risks. Another avenue towards safe super powerful AI is that they are designed to be corrigible [144] and therefore can be corrected and modified.

Appendix B

Proposal: Aligning Large Language Models to Publicly Collected Human Values

B.1 Research aims and questions

The goal of this project is to address some gaps in the current AI alignment landscape by a proof-of-concept end-to-end project. It will attempt to align a large language model to a set of human values collected from the public and to develop an evaluation method that can better measure alignment success on these values. To do this I will use value survey data to do alignment training and evaluate using behavioral assessments of the LLM by the public. Specifically, the project will set out to answer a series of research questions.

RQ1: **Alignment feasibility.** Do simple post-training methods (e.g. SFT) work to appreciably align a large language model to a set of human values?

RQ2: **Evaluation.** What method can be used to evaluate the alignment of a large language model to a set of representatively collected human values, in a way that remains democratically grounded?

These research questions, if addressed, would provide useful insights into the feasibility of aligning large language models to the public in a democratically grounded way[1]. This is of particular importance within the context of Public, Sovereign and Democratic AI [47, 104, 118, 110] movements that are pushing for more public participation, oversight, and accountability in AI systems.

This project sets out with some hypotheses around these research questions.

H1: A small (<40B parameters) **open-weights** LLM has the ability to understand the survey questions and model a respondents values when given its responses.
Justification: Small LLMs have been shown to respond positively to fine tuning with value/opinion survey data with the result of better prediction of survey responses [148].

H2: Fine-tuning a base LLM by simply telling it which answers to a survey question is "correct" will change its behavior in downstream tasks.
Justification: Recent work [105] has suggested that fine tuning methods similar to this has produced detectable behavioral differences in the LLM in sensitive situations.

[1]These research questions are supported by efforts like [6]

H3: The questions and responses in the WVS are useful to predict behavior, as tested by asking respondents how they would act in certain situations.
Justification: The foundations of these survey methods is that they allow us to understand humans values, many years of research has suggested that values are good predictors of behaviour [139, 125].

H4: The public will rate models that are fine-tuned to their survey responses as most aligned to them compared with models fine-tuned to other clusters or the baseline model.
Justification: Individuals are shown to prefer agents that they perceive share common values with them which is called similarity-attraction [32].

B.2 Alignment methodology and evaluation plan

To address these research questions, I will need to carry out a sequence of steps from ingesting a pre-trained model to outputting an aligned model and evaluation results.

B.2.1 Defining the alignment target

For pragmatic reasons, I will use existing survey data from the World Values Survey to define a set of human values to align towards. This will provide me with a dataset of 300 questions answered by about 1,000 respondents in multiple countries. Either each country will be treated as a separate cluster or I will cluster respondents based on their answers to define value clusters. This will allow me to explore pluralistic alignment targets.

I will define two sorts of alignment targets. The first will be a set of global values that represent the values of the population as a whole, this can be derived by taking the mode of the entire response along with a variation metric. The second will be pluralistic targets that get defined as the mode response within each cluster of respondents (one cluster for each country). This will allow me to explore the differences between different value alignment targets.

B.2.2 Aligning to the targets

Recent work [105] suggests that simple SFT may have appreciable alignment effects. Therefore, I will attempt to use SFT by getting the model to answer the survey questions correctly, in Figure B.1 one can see the basic outline. If this proves insufficient, I will explore more complex methods such as KTO with binary feedback (correct survey response, incorrect survey response). These methods will be used to align a base LLM to the defined alignment targets from step 1.

Given there are only about 290 survey questions, I will need to augment the data to ensure there are enough training examples. I will do this by using different templates that pose the same question in different ways. Importantly the actual content of the question will remain the same, only the system prompt and structure of getting the LLM to answer will change. By having a set of different templates it is plausible that the 290 initial questions can be augmented to create a training set of about 1500 examples.

B.2.3 Evaluating the success of the alignment

To evaluate the success of the alignment, I will need to develop an evaluation method that can measure how well the AI agent is aligned to the defined human values. There will be broadly two tests. One is automated evaluation using held-out survey questions. The other

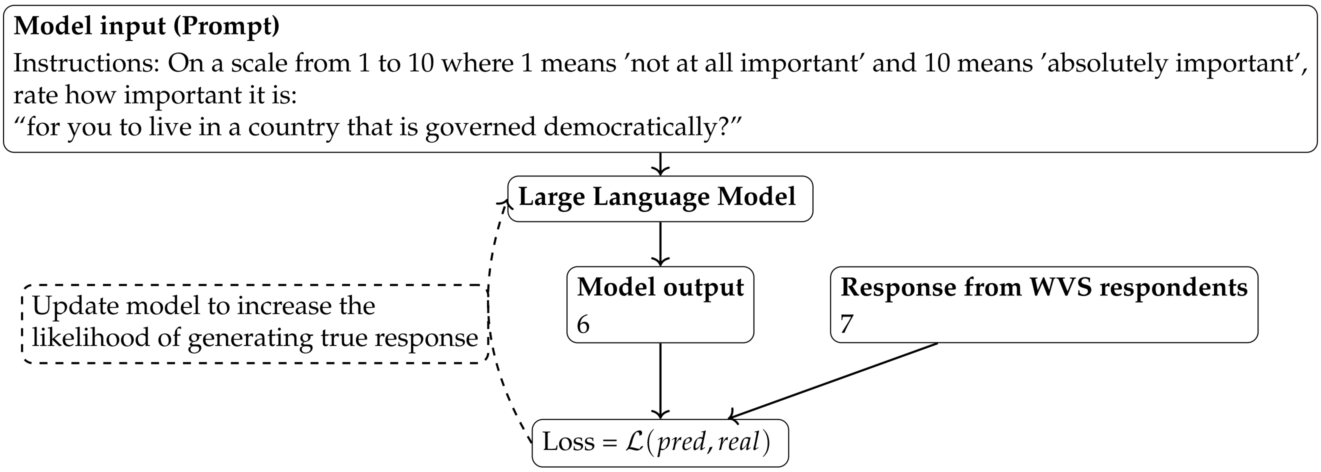

Figure B.1: Simplified illustrative example using WVS survey data. The model is prompted with a survey question, produces a Likert-scale response, and receives a supervision signal based on its distance from the observed WVS response.

is a public consultation process where I will collect feedback on how much the person feels the model acted in alignment with their values. This will require significant design to ensure that the consultation is representative and captures the necessary feedback.

The first evaluation method will involve holding out a set of survey questions from the training data. The test split will be done by splitting out questions that are semantically similar to each other and holding out different question templates. This will ensure that the model has at least some ability to generalize to unseen questions.

The second evaluation method will involve getting the model to engage in various simulations of real world tasks. Then I will recruit participants from the various clusters defined in step 1 to rank which model actions best align with their values in these scenarios. This will provide a human-grounded evaluation of alignment success.

These evaluations combined will provide a robust measure of how well the model has understood the human values as well as how it actually affects its behavior in more realistic settings.

B.3 Rationale for the project

Current methods for training large language models involve scraping the Internet for large amounts of text data, creating powerful next-token prediction models. To turn these into something useful for end users (e.g. chatbots or assistants), the model is then aligned to behave in desired ways, typically by contracting human labelers/annotators/demonstrators to provide training data. This has made modern LLMs highly useful. Problematically, current alignment methods are opaque and the alignment target is often specified privately by a small set of individuals (e.g. the people writing instructions for annotators). This requires trust from the public that the model is aligned with their values. Even if the values of the creators were initially acceptable, they are usually set within for-profit organizations with incentives that can distort alignment (e.g. maximizing engagement or revenue). Therefore we cannot assume these models are aligned to the public good.

There are multiple challenges, however, with aligning models to the public good, or more formally maximizing median human flourishing. First is how to define what the public good is, that is what would the alignment target even look like? Avenues to explore are; lists of principles (like constitutional AI [13]), value sets from a taxonomy of human values [136, 125], fine grained preference datasets (like that collected for RLHF [109] or similar), or more complex structures like Moral graphs [81]. Second is that even if we understood what an alignment target should look like, it is non-obvious how to collect data that represents it in a way that is representative of the public. Third is that once this alignment structure is defined and data collected to turn it into a valid instance of an alignment target it is an open question how best to align a subsequent model towards this target. This becomes particularly challenging given that the alignment target could be a pluralistic target and thus require fair distinct treatment of different subgroups. Finally, even if we had a method to align a model to this target, it is unclear how we would evaluate success. Current methods are generally short term and often take the form of pairwise comparisons or held-out test sets of expected responses. Notably, they do not usually provide out-of-distribution evaluation or capture multi-step interactions and downstream outcomes, and they are often designed and performed by the same select few individuals who designed the alignment target in the first place.

This project aims to address these challenges by providing a demonstration that we could use readily available value survey data to define an alignment target. Then train a model to align to this target using simple methods. Finally, evaluation of alignment in out-of-distribution scenarios can be done by judging the model's behavior in realistic scenarios by members of the public.

B.4 Contributions

The project overall will contribute by providing a proof-of-concept alignment procedure to the NZ public. Within that it will contribute:

- A method to define alignment targets and a training dataset from survey data that captures pluralistic human values in a democratically defensible manner.
- A novel [application of][2] evaluation dataset to test a models behavior in realistic scenarios to be judged by human participants from various value clusters.
- A demonstration of a lightweight training procedure on open weight LLMs that has democratically fair evaluation across different sub groups.[3]

All of the work will be open-sourced, documented and freely available.

B.5 Resources and feasibility

This project will take just over 4 months to complete and require a few thousand dollars of compute and participant compensation.

B.5.1 Proposed timeline

Figure B.2 shows the proposed timeline for a 18-week project.

[2] Unsure if creating my own or using someone else's dataset

[3] Important to note I don't know how successful this will be yet. Ideally I could say that it achieves high alignment scores but this is unknown at this stage.

B.5.2 Computational resources

To train and evaluate the model, hardware will be needed. If I can train models with about 30 billion parameters, I will be able to get near-SOTA performance [4]. Using modern quantization methods [52, 51] and parameter-efficient fine-tuning methods [69, 52], I should be able to fine-tune these models using around 80GB of VRAM (h100, h200 etc). Therefore, I estimate needing about 240 USD of GPU time to carry out the fine-tuning and evaluation if hardware is rented in the cloud [5]. Alternatively the university has some GPU resources that could be used and might be sufficient.

B.5.3 Survey data

The initial alignment target will be defined using existing survey data from the World Values Survey (WVS) [64] survey data from across the globe with about 1,000 respondents in each country. It is openly accessible for research purposes [6].

B.5.4 Public consultation responses

To create a rigorous evaluation of the alignment process a public consultation process will be needed. This will require recruiting participants from various clusters of the NZ population to provide feedback. Each participants will rate how much an AI agents actions (i.e responses) align with their values. I would expect to need at least 50 people from various countries to complete about a 30 minutes task (including instructions). The complete design of this is left to the research phase.

B.6 Ethical considerations

The interaction of this project with human data and human participants raises ethical concerns. Furthermore as AI has the potential for strong negative real-world impact there are considerations around development and training of these models.

- The use of human survey data (WVS) to construct alignment targets.
  mitigation: Only aggregate-level data will be used, and all data custodians' access and usage conditions will be followed.
- Interaction with human participants during the public consultation evaluation.
  mitigation: Ethics approval will be sought prior to recruitment, and informed consent will be obtained from all participants.
- Concerns of dual-use and capability increases from alignment work.
  mitigation: The project will use existing open-weight models (not frontier-scale training) and focus on value alignment rather than capability enhancement. Concerns of emergent misalignment will be addressed by only training to ever do "positive" things.

[4] Based on the Artificial Analysis leaderboards: 30B parameters achieve about 60% of the performance of frontier models.

[5] Each fine-tuning run will take in the realm of 3–6 hours depending on the size of the model, amount of data used, alignment methods used, and GPU hardware. About 80 hours (5 hours per fine-tune * 3 different methods * 5 different value sets + 5 hours of evaluation) of GPU time will be needed at a cost of 2–3 USD for a single hour on a high-end GPU (e.g. H200) [162]

[6] Initial plans of using NZAVS data was not possible due to failure to obtain access

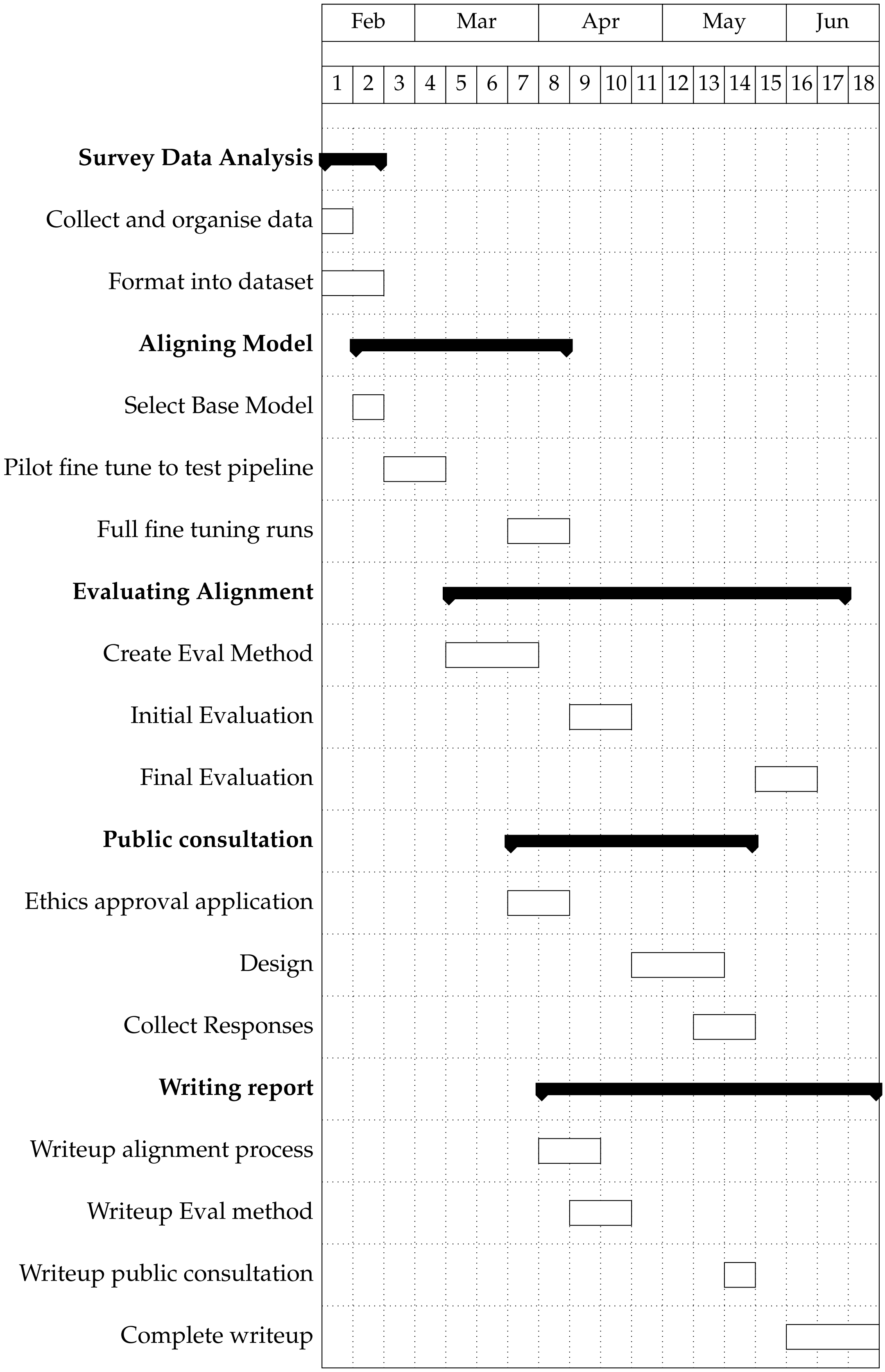

Figure B.2: Project timeline in weeks (only roughly aligned to months). This assumes a start time of 9th of February 2026.